

TECHNICAL MATHEMATICS



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Technical Mathematics

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Licensing

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CHAPTER OVERVIEW

1: The Whole Numbers

Welcome to the study of prealgebra. In this first chapter of study, we will introduce the set of natural numbers, then follow with the set of whole numbers. We will then follow with a quick review of addition, subtraction, multiplication, and division skills involving whole numbers that are prerequisite for success in the study of prealgebra. Along the way we will introduce a number of properties of the whole numbers and show how that can be used to evaluate expressions involving whole number operations. We will also define what is meant by *prime* and *composite* numbers, discuss a number of divisibility tests, then show how any composite number can be written uniquely as a product of prime numbers. This will lay the foundation for requisite skills with fractional numbers in later chapters. Finally, we will introduce the concept of a *variable*, then introduce equations and technique required for their solution. We will use equations to model and solve a number of real-world applications along the way.

Let's begin the journey.

- [1.1: An Introduction to the Whole Numbers](#)
- [1.2: Adding and Subtracting Whole Numbers](#)
- [1.3: Multiplication and Division of Whole Numbers](#)
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1.1: An Introduction to the Whole Numbers

A *set* is a collection of objects. If the set is finite, we can describe the set completely by simply listing all the objects in the set and enclosing the list in curly braces. For example, the set

$$S = \{dog, cat, parakeet\}$$

is the set whose members are “dog”, “cat”, and “parakeet.” If the set is infinite, then we need to be more clever with our description. For example, the set of *natural numbers* (or *counting numbers*) is the set

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}.$$

Because this set is infinite (there are an infinite number of natural numbers), we can’t list all of them. Instead, we list the first few then follow with “three dots,” which essentially mean “etcetera.” The implication is that the reader sees the intended pattern and can then intuit the remaining numbers in the set. Can you see that the next few numbers are 6, 7, 8, 9, etc.?

If we add the number zero to the set of natural numbers, then we have a set of numbers that are called the *whole numbers*.

The Whole Numbers

The set

$$\mathbb{W} = \{0, 1, 2, 3, 4, 5, \dots\}$$

is called the set of *whole numbers*.

The whole numbers will be our focus in the remainder of this chapter.

Graphing Numbers on the Number Line

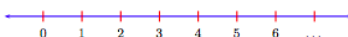
It is a simple matter to set up a correspondence between the whole numbers and points on a **number line**. First, draw a number line, then set a tick mark at zero.



The next step is to declare a unit length.



The remainder of the whole numbers now fall easily in place on the number line.



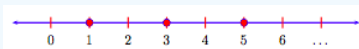
When asked to graph a whole number on a number line, shade in a solid dot at the position on the number line that corresponds to the given whole number.

✓ Example 1.

Graph the whole numbers 1, 3, and 5 on the number line.

Solution

Shade the numbers 1, 3, and 5 on the number line as solid dots.



? Exercise

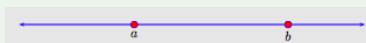
Graph the whole numbers 3, 4, and 6 on the number line.

Ordering the whole numbers

Now that we have a correspondence between the whole numbers and points on the number line, we can order the whole numbers in a natural way. Note that as you move to the left along the number line, the numbers get smaller; as you move to the right, the numbers get bigger. This inspires the following definition.

Ordering the Whole Numbers

Suppose that a and b are whole numbers located on the number line so that the point representing the whole number a lies to the left of the point representing the whole number b .



Then the whole number a is “less than” the whole number b and write

$$a < b.$$

Alternatively, we can also say that the whole number b is “greater than” the whole number a and write

$$b > a.$$

Comparison Property

When comparing two whole numbers a and b , only one of three possibilities is true:

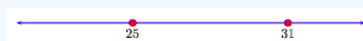
$$a < b \text{ or } a = b \text{ or } a > b.$$

✓ Example 2.

Compare the whole numbers 25 and 31.

Solution

On the number line, 25 is located to the left of 31.



Therefore, 25 is less than 31 and write $25 < 31$. Alternatively, we could also note that 31 is located to the right of 25. Therefore, 31 is greater than 25 and write $31 > 25$.

? Exercise

Compare the whole numbers 18 and 12.

Answer

$$18 > 12.$$

Expanded Notation

The whole numbers

$$D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

are called *digits* and are used to construct larger whole numbers. For example, consider the whole number 222 (pronounced “two hundred twenty two”). It is made up of three twos, but the position of each two describes a different meaning or value.

2	2	2
hundreds	tens	ones

- The first two is in the “hundreds” position and represents two hundreds or 200.

- The second two is in the “tens” position and represents two tens or 20.
- The third two is in the “ones” position and represents two ones or 2.

Consider the larger number 123,456,789. The following table shows the place value of each digit.

1	2	3	4	5	6	7	8	9
hundred millions	ten millions	millions	hundred thousands	ten thousands	thousands	hundreds	tens	ones
millions			thousands			ones		

In “expanded notation,” we would write

1 hundred million + 2 ten millions + 3 millions + 4 hundred thousands + 5 ten thousands + 6 thousands + 7 hundreds + 8 tens + 9 ones.

We read the numeral 123,456,789 as “one hundred twenty three million, four hundred fifty six thousand, seven hundred eighty nine.”

Let’s look at another example.

✓ Example 3.

Write the number 23,712 in expanded notation, then pronounce the result.

Solution

In expanded notation, 23,712 becomes

2 ten thousands + 3 thousands + 7 hundreds + 1 ten + 2 ones.

This is pronounced “twenty three thousand, seven hundred twelve.”

? Exercise

Write the number 54,615 in expanded notation. Pronounce the result.

✓ Example 4.

Write the number 203,405 in expanded notation, then pronounce the result.

Solution

In expanded notation, 203,405 becomes

2 hundred thousands + 0 ten thousands + 3 thousands + 4 hundreds + 0 tens + 5 ones.

Since 0 ten thousands is zero and 0 tens is also zero, this can also be written

2 hundred thousands + 3 thousands + 4 hundreds + 5 ones.

This is pronounced “two hundred three thousand, four hundred five.”

? Exercise

Write the number 430,705 in expanded notation. Pronounce the result.

Answer

9,440

Rounding whole numbers

When less precision is needed, we round numbers to a particular place. For example, suppose a store owner needs approximately 87 boxes of ten-penny nails, but they can only be ordered in cartons containing ten boxes.



Note that 87 is located closer to 9 tens (or 90) than it is to 8 tens (or 80). Thus, rounded to the nearest ten, $87 \approx 90$ (87 approximately equals 90). The store owner decides that 90 boxes is probably a better fit for his needs. On the other hand, the same store owner estimates that he will need 230 bags of peatmoss for his garden section.



Note that 230 is closer to 2 hundreds (or 200) than it is to 3 hundreds (or 300). The store owner worries that might have overestimated his need, so he rounds down to the nearest hundred, $230 \approx 200$ (230 approximately equals 200). There is a simple set of rules to follow when rounding.

Rules for Rounding

To round a number to a particular place, follow these steps:

1. Mark the place you wish to round to. This is called the rounding digit.
2. Check the next digit to the right of your digit marked in step 1. This is called the test digit.
 - a) If the test digit is greater than or equal to 5, add 1 to the rounding digit and replace all digits to the right of the rounding digit with zeros.
 - b) If the test digit is less than 5, keep the rounding digit the same and replace all digits to the right of rounding digit with zero

Let's try these rules with an example or two.

✓ Example 5

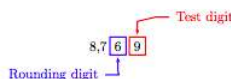
Round the number 8,769 to the nearest ten.

Solution

Mark the rounding and test digits.

? Exercise

Round the number 9,443 to the nearest ten.



The test digit is greater than 5. The "Rules for Rounding" require that we add 1 to the rounding digit, then make all digits to the right of the rounding digit zeros. Thus, rounded to the nearest ten,

$$8,769 \approx 8,770.$$

That is, 8,769 is *approximately* equal to 8,770.

Mathematical Notation.

The symbol

$\approx$

means *approximately equal*.

✓ Example 6.

Round the number 4,734 to the nearest hundred.

Solution

Mark the rounding and test digits.



The test digit is less than 5. The “Rules for Rounding” require that we keep the rounding digit the same, then make all digits to the right of the rounding digit zeros. Thus, rounded to the nearest hundred,

$$4,734 \approx 4,700.$$

? Exercise

Round the number 6,656 to the nearest hundred.

Answer

6,700

Table 1.1: Atmospheric CO₂ values (ppmv) derived from in situ air samples collected at Mauna Loa, Hawaii, USA.

Year	1965	1975	1985	1995	2005
Atmospheric CO ₂	319	330	344	359	378

Tables and Graphs

Reading data in graphical form is an important skill. The data in Table 1.1 provides measures of the carbon dioxide content (CO₂) in the atmosphere, gathered in the month of January at the observatory atop Mauna Loa in Hawaii.

In Figure 1.1(a), a *bar graph* is used to display the carbon dioxide measurements. The year the measurement was taken is placed on the horizontal axis, and the height of each bar equals the amount of carbon dioxide in the atmosphere during that year.

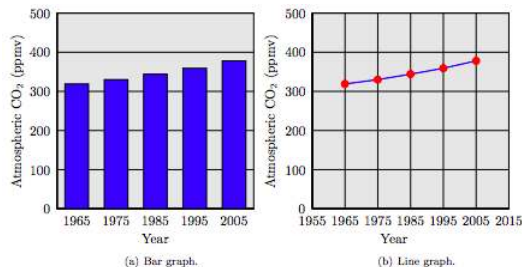


Figure 1.1: Using graphs to examine carbon dioxide data.

In Figure 1.1(b), a *line graph* is used to display the carbon dioxide measurements. Again, the dates of measurement are placed on the horizontal axis, and the amount of carbon dioxide in the atmosphere is placed on the vertical axis. Instead of using the height of a bar to represent the carbon dioxide measurement, we place a dot at a height that represents the carbon monoxide content. Once each data point is plotted, we connect consecutive data points with line segments.

Exercises

In Exercises 1-12, sketch the given whole numbers on a number line, then arrange them in order, from smallest to largest.

1. 2, 8, and 4
2. 2, 7, and 4
3. 1, 8, and 2
4. 0, 4, and 3
5. 0, 4, and 1
6. 3, 6, and 5
7. 4, 9, and 6
8. 2, 4, and 3
9. 0, 7, and 4
10. 2, 8, and 6
11. 1, 6, and 5
12. 0, 9, and 5

In Exercises 13-24, create a number line diagram to determine which of the two given statements is true.

13. $3 < 8$ or $3 > 8$
14. $44 < 80$ or $44 > 80$
15. $59 < 24$ or $59 > 24$
16. $15 < 11$ or $15 > 11$
17. $0 < 74$ or $0 > 74$
18. $11 < 18$ or $11 > 18$
19. $1 < 81$ or $1 > 81$
20. $65 < 83$ or $65 > 83$
21. $43 < 1$ or $43 > 1$
22. $62 < 2$ or $62 > 2$
23. $43 < 28$ or $43 > 28$
24. $73 < 21$ or $73 > 21$

-
25. Which digit is in the thousands column of the number 2,054,867,372?
 26. Which digit is in the hundreds column of the number 2,318,999,087?
 27. Which digit is in the hundred thousands column of the number 8,311,900,272?
 28. Which digit is in the tens column of the number 1,143,676,212?
 29. Which digit is in the hundred millions column of the number 9,482,616,000?
 30. Which digit is in the hundreds column of the number 375,518,067?
 31. Which digit is in the ten millions column of the number 5,840,596,473?
 32. Which digit is in the hundred thousands column of the number 6,125,412,255?
 33. Which digit is in the hundred millions column of the number 5,577,422,501?
 34. Which digit is in the thousands column of the number 8,884,966,835?

35. Which digit is in the tens column of the number 2,461,717,362?

36. Which digit is in the ten millions column of the number 9,672,482,548?

37. Round the number 93, 857 to the nearest thousand.

38. Round the number 56, 872 to the nearest thousand.

39. Round the number 9, 725 to the nearest ten.

40. Round the number 6, 815 to the nearest ten.

41. Round the number 58, 739 to the nearest hundred.

42. Round the number 93, 146 to the nearest hundred.

43. Round the number 2, 358 to the nearest ten.

44. Round the number 8, 957 to the nearest ten.

45. Round the number 39, 756 to the nearest thousand.

46. Round the number 24, 965 to the nearest thousand.

47. Round the number 5, 894 to the nearest ten.

48. Round the number 3, 281 to the nearest ten.

49. Round the number 56, 123 to the nearest hundred.

50. Round the number 49, 635 to the nearest hundred.

51. Round the number 5, 483 to the nearest ten.

52. Round the number 9, 862 to the nearest ten.

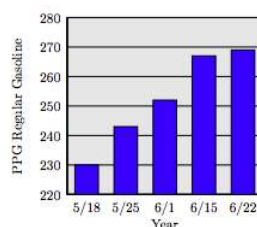
53. According to the U.S. Census Bureau, the estimated population of the US is 304,059,724 as of July 2008. Round to the nearest hundred thousand.

54. According to the U.S. Census Bureau, the estimated population of California is 36,756,666 as of July 2008. Round to the nearest hundred thousand.

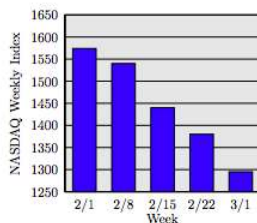
55. According to the U.S. Census Bureau, the estimated population of Humboldt County is 129,000 as of July 2008. Round to the nearest ten thousand.

56. According to the U.S. Census Bureau, the estimated population of the state of Alaska was 686,293 as of July 2008. Round to the nearest ten thousand.

57. The following bar chart shows the average price (in cents) of one gallon of regular gasoline in the United States over five consecutive weeks in 2009, running from May 18 (5/18) through June 22 (6/22). What was the price (in cents) of one gallon of regular gasoline on June 1, 2009?



58. The following bar chart shows the average weekly NASDAQ index for five consecutive weeks in 2009, beginning with week starting February 1 (2/1) and ending with the week starting March 1 (3/1). What was the average NASDAQ index for the week starting February 8, 2009?



59. The population of Humboldt County is broken into age brackets in the following table. *Source: WolframAlpha.*

Age in years	Number
under 5	7,322
5 – 18	26,672
18 – 65	78,142
over 65	16,194

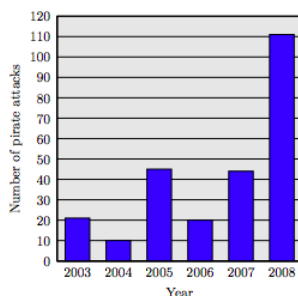
Create a bar chart for this data set with one bar for each age category.

60. The five cities with the largest number of reported violent crimes in the year 2007 are reported in the following table. *Source: Wikipedia.*

City	Violent Crimes
Setroit	2,289
St. Louis	2,196
Memphis	1,951
Oakland	1,918
Baltimore	1,631

Create a bar chart for this data set with one bar for each city.

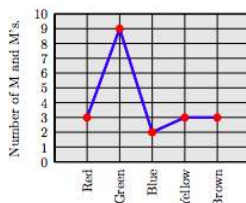
61. The following bar chart tracks pirate attacks off the coast of Somalia.



Source: ICC International Maritime Bureau, AP Times-Standard, 4/15/2009

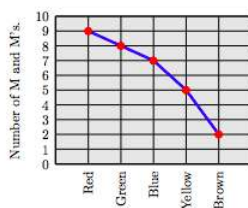
- How many pirate attacks were there in 2003?
- How many pirate attacks were there in 2008?

62. A team of students separated a small bowl of M and M's into five piles by color. The following line plot indicates the number of M and M's of each color.



How many red M and M's were in the bowl?

63. A team of students separated a small bowl of M and M's into five piles by color. The following line plot indicates the number of M and M's of each color.



How many red M and M's were in the bowl?

64. A team of students separated a small bowl of M and M's into five piles by color. The following table indicates the number of M and M's of each color.

Color	Number
Red	5
Green	9
Blue	7
Yellow	2
Brown	3

Create a lineplot for the M and M data. On the horizontal axis, arrange the colors in the same order as presented in the table above.

65. A team of students separated a small bowl of M and M's into five piles by color. The following table indicates the number of M and M's of every color.

Color	Number
Red	3
Green	7
Blue	2
Yellow	4
Brown	9

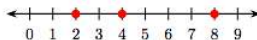
Create a lineplot for the M and M data. On the horizontal axis, arrange the colors in the same order as presented in the table above.

66. *Salmon count.* The table shows the number of adult coho salmon returning to the Shasta River over the past four years. Round the salmon count for each year to the nearest ten. *Times-Standard Shasta River coho rescue underway.*

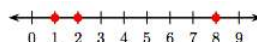
Year	Salmon count
2007	300
2008	31
2009	9
2010	4

Answers

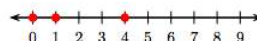
1. Smallest to largest: 2, 4, and 8.



3. Smallest to largest: 1, 2, and 8.



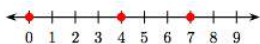
5. Smallest to largest: 0, 1, and 4.



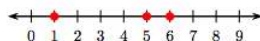
7. Smallest to largest: 4, 6, and 9.



9. Smallest to largest: 0, 4, and 7.



11. Smallest to largest: 1, 5, and 6.



13.



Therefore, $3 < 8$.

15.



Therefore, $59 > 24$.

17.



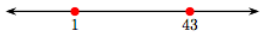
Therefore, $0 < 74$.

19.



Therefore, $1 < 81$.

21.



Therefore, $43 > 1$.

23.



Therefore, $43 > 28$.

25. 7

27. 9

29. 4

31. 4

33. 5

35. 6

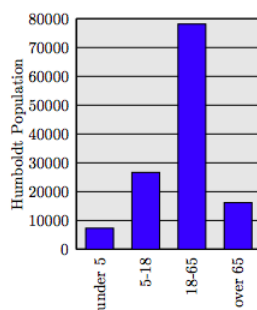
37. 94000

39. 9730

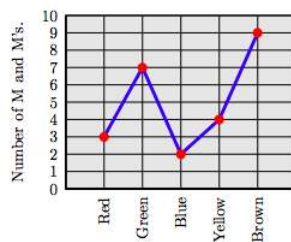
41. 58700

43. 2360

- 45. 40000
- 47. 5890
- 49. 56100
- 51. 5480
- 53. 304,100,000
- 55. 130,000
- 57. Approximately 252 cents
- 59.



- 61.
 - a) Approximately 21
 - b) Approximately 111
- 63. 9
- 65.



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1.2: Adding and Subtracting Whole Numbers

In the expression $3 + 4$, which shows the sum of two whole numbers, the whole numbers 3 and 4 are called *addends* or *terms*. We can use a visual approach to find the sum of 3 and 4. First, construct a number line as shown in Figure 1.2.

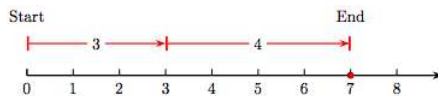


Figure 1.2: Adding whole numbers on the number line.

To add 3 and 4, proceed as follows.

1. Start at the number 0, then draw an arrow 3 units to the right, as shown in Figure 1.2. This arrow has magnitude (length) three and represents the whole number 3.
2. Draw a second arrow of length four, starting at the end of the first arrow representing the number 3. This arrow has magnitude (length) four and represents the whole number 4.
3. The sum of 3 and 4 could be represented by an arrow that starts at the number 0 and ends at the number 7. However, we prefer to mark this sum on the number line as a solid dot at the whole number 7. This number represents the sum of the whole numbers 3 and 4.

The Commutative Property of Addition

Let's change the order in which we add the whole numbers 3 and 4. That is, let's find the sum $4 + 3$ instead.



Figure 1.3: Addition is commutative; i.e., order doesn't matter.

As you can see in Figure 1.3, we start at zero then draw an arrow of length four, followed by an arrow of length three. However, the result is the same; i.e., $4 + 3 = 7$.

Thus, the order in which we add three and four does not matter; that is,

$$3 + 4 = 4 + 3.$$

This property of addition of whole numbers is known as the *commutative property* of addition.

The Commutative Property of Addition

Let a and b represent two whole numbers. Then,

$$a + b = b + a.$$

Grouping Symbols

In mathematics, we use *grouping symbols* to affect the order in which an expression is evaluated. Whether we use parentheses, brackets, or curly braces, the expression inside any pair of grouping symbols must be evaluated first. For example, note how we first evaluate the sum in the parentheses in the following calculation.

$$\begin{aligned}(3 + 4) + 5 &= 7 + 5 \\ &= 12\end{aligned}$$

The rule is simple: Whatever is inside the parentheses is evaluated first.

Writing Mathematics

When writing mathematical statements, follow the mantra:

One equal sign per line.

We can use brackets instead of parentheses.

$$\begin{aligned}5 + [7 + 9] &= 5 + 16 \\ &= 21\end{aligned}$$

Again, note how the expression inside the brackets is evaluated first.

We can also use curly braces instead of parentheses or brackets.

$$\begin{aligned}\{2+3\} + 4 &= 5 + 4 \\ &= 9\end{aligned}$$

Again, note how the expression inside the curly braces is evaluated first.

If grouping symbols are nested, we evaluate the innermost parentheses first.

For example,

$$\begin{aligned}2 + [3 + (4 + 5)] &= 2 + [3 + 9] \\ &= 2 + 12 \\ &= 14.\end{aligned}$$

Grouping Symbols

Use parentheses, brackets, or curly braces to delimit the part of an expression you want evaluated first. If grouping symbols are nested, evaluate the expression in the innermost pair of grouping symbols first.

The Associative Property of Addition

Consider the evaluation of the expression $(2+3)+4$. We evaluate the expression in parentheses first.

$$\begin{aligned}(2 + 3) + 4 &= 5 + 4 \\ &= 9\end{aligned}$$

Now, suppose we change the order of addition to $2 + (3 + 4)$. Then,

$$\begin{aligned}2 + (3 + 4) &= 2 + 7 \\ &= 9.\end{aligned}$$

Although the grouping has changed, the result is the same. That is,

$$(2 + 3) + 4 = 2 + (3 + 4).$$

This property of addition of whole numbers is called the *associate property* of addition.

Associate Property of Addition

Let a , b , and c represent whole numbers. Then,

$$(a + b) + c = a + (b + c).$$

Because of the associate property of addition, when presented with a sum of three numbers, whether you start by adding the first two numbers or the last two numbers, the resulting sum is the same.

The Additive Identity

Imagine a number line visualization of the sum of four and zero; i.e., $4 + 0$.

In Figure 1.4, we start at zero, then draw an arrow of magnitude (length) four pointing to the right. Now, at the end of this arrow, attach a second arrow of length zero. Of course, that means that we remain right where we are, at 4. Hence the shaded dot at 4 is the sum. That is, $4 + 0 = 4$.

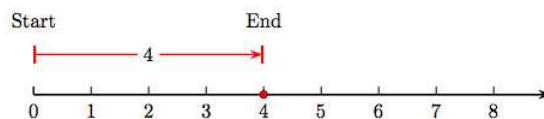


Figure 1.4: Adding zero to four.

The Additive Identity Property

The whole number zero is called the additive identity. If a is any whole number, the

$$a + 0 = a.$$

The number zero is called the additive identity because if you add zero to any number, you get the identical number back.

Adding Larger Whole Numbers

For completeness, we include two examples of adding larger whole numbers. Hopefully, the algorithm is familiar from previous coursework.

✓ Example 1

Simplify: $1,234 + 498$.

Solution

Align the numbers vertically, then add, starting at the furthest column to the right. Add the digits in the ones column, $4 + 8 = 12$. Write the 2, then carry a 1 to the tens column. Next, add the digits in the tens column, $3 + 9 = 12$, add the carry to get 13, then write the 3 and carry a 1 to the hundreds column. Continue in this manner, working from right to left

$$\begin{array}{r} 11 \\ 1234 \\ + 498 \\ \hline 1732 \end{array}$$

Therefore, $1,234 + 498 = 1,732$

? Exercise

Simplify: $1,286 + 349$.

Answer

1635

Add three or more numbers in the same manner.

✓ Example 2

Simplify: $256 + 322 + 418$.

Solution

Align the numbers vertically, then add, starting at the furthest column to the right. Add the digits in the ones column, $6 + 2 + 8 = 16$. Write the 6, then carry a 1 to the tens column. Continue in this manner, working from right to left.

$$\begin{array}{r} 256 \\ 322 \\ + 418 \\ \hline 996 \end{array}$$

Therefore, $256 + 322 + 418 = 996$.

? Exercise

Simplify: $256 + 342 + 283$

Answer

881

Subtraction of Whole Numbers

The key idea is this: *Subtraction is the opposite of addition.* For example, consider the difference $7 - 4$ depicted on the number line in Figure 1.5.

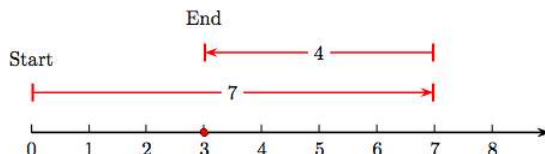


Figure 1.5: Subtraction means *add the opposite*.

If we were adding 7 and 4, we first draw an arrow starting at zero pointing to the right with magnitude (length) seven. Then, to add 4, we would draw a second arrow of magnitude (length) 4, attached to the end of the first arrow and pointing to the right.

However, because subtraction is the *opposite* of addition, in Figure 1.5 we attach an arrow of magnitude (length) four to the end of the first arrow, but *pointing in the opposite direction* (to the left). Note that this last arrow ends at the answer, which is a shaded dot on the number line at 3. That is, $7 - 4 = 3$.

Note that subtraction is **not commutative**; that is, it make no sense to say that $7 - 5$ is the same as $5 - 7$.

Subtraction is **not associative**. It is not the case that $(9 - 5) - 2$ is the same as $9 - (5 - 2)$. On the one hand,

$$\begin{aligned}(9 - 5) - 2 &= 4 - 2 \\ &= 2,\end{aligned}$$

but

$$\begin{aligned}9 - (5 - 2) &= 9 - 3 \\ &= 6.\end{aligned}$$

Subtracting Larger Whole Numbers

Much as we did with adding larger whole numbers, to subtract two large whole numbers, align them vertically then subtract, working from right to left. You may have to “borrow” to complete the subtraction at any step.

✓ Example 3

Simplify: $1,755 - 328$.

Solution

Align the numbers vertically, then subtract, starting at the ones column, then working right to left. At the ones column, we cannot subtract 8 from 5, so we borrow from the previous column. Now, 8 from 15 is 7. Continue in this manner, working from right to left.

$$\begin{array}{r} \overset{4}{\cancel{7}} \overset{1}{\cancel{5}} \\ - 328 \\ \hline 1427 \end{array}$$

Therefore, $1,755 - 328 = 1,427$.

? Exercise

Simplify: $5,635 - 288$.

Answer

5,347

Order of Operations

In the absence of grouping symbols, it is important to understand that addition holds no precedence over subtraction, and vice-versa.

Perform all additions and subtractions in the order presented, moving left to right.

Let's look at an example.

✓ Example 4

Simplify the expression $15 - 8 + 4$.

Solution

This example can be trickier than it seems. However, if we follow the rule (perform all additions and subtractions in the order presented, moving left to right), we should have no trouble. First comes fifteen minus eight, which is seven. Then seven plus four is eleven.

$$\begin{aligned} 15 - 8 + 4 &= 7 + 4 \\ &= 11. \end{aligned}$$

? Exercise

Simplify: $25 - 10 + 8$.

Answer

23

⚠ Caution! Incorrect answer ahead!

Note that it is possible to arrive at a different (but incorrect) answer if we favor addition over subtraction in Example 4. If we first add eight and four, then $15 - 8 + 4$ becomes $15 - 12$, which is 3. However, note that **this is incorrect**, because it violates the rule “perform all additions and subtractions in the order presented, moving left to right.”

Applications — Geometry

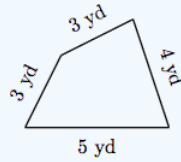
There are any number of applications that require a sum or difference of whole numbers. Let's examine a few from the world of geometry.

Perimeter of a Polygon

In geometry a polygon is a plane figure made up of a closed path of a finite sequence of segments. The segments are called the *edges* or *sides* of the polygon and the points where two edges meet are called the *vertices* of the polygon. The *perimeter* of any polygon is the sum of the lengths of its sides.

✓ Example 5

A quadrilateral is a polygon with four sides. Find the perimeter of the quadrilateral shown below, where the sides are measured below.



Solution

To find the perimeter of the quadrilateral, find the sum of the lengths of the sides.

$$\text{Perimeter} = 3 + 3 + 4 + 5 = 15$$

Hence, the perimeter of the quadrilateral is 15 yards.

? Exercise

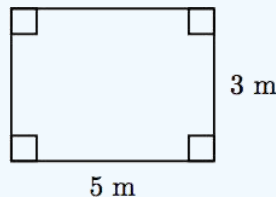
A quadrilateral has sides that measure in 4 in., 3 in., 5 in., and 5 in. Find the perimeter.

Answer

17 inches

✓ Example 6

A quadrilateral (four sides) is a rectangle if all four of its angles are right angles. It can be shown that the opposite sides of a rectangle must be equal. Find the perimeter of the rectangle shown below, where the sides of the rectangle are measured in meters.



Solution

To find the perimeter of the rectangle, find the sum of the four sides. Because opposite sides have the same length, we have two sides of length 5 meters and two sides of length 3 meters. Hence,

$$\text{Perimeter} = 5 + 3 + 5 + 3 = 16.$$

Thus, the perimeter of the rectangle is 16 meters.

? Exercise

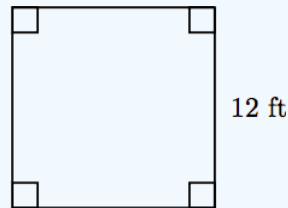
A rectangle has length 12 meters and width 8 meters. Find its perimeter.

Answer

40 meters

✓ Example 7

A quadrilateral (four sides) is a *square* if all four of its sides measures 18 centimeters. Find its perimeter. are equal and all four of its angles are right angles. Pictured below is a square having a side of length 12 feet. Find the perimeter of the square.



Solution

Because the quadrilateral is a square, all four sides have the same length, namely 12 feet. To find the perimeter of the square, find the sum of the four sides. Perimeter = $12 + 12 + 12 + 12 = 48$

Hence, the perimeter of the square is 48 feet.

? Exercise

A square has a side that measures 18 centimeters. Find its perimeter.

Answer

72 centimeters

Application — Alternative Fuels

Automobiles that run on alternative fuels (other than gasoline) have increased in the United States over the years.

✓ Example 8

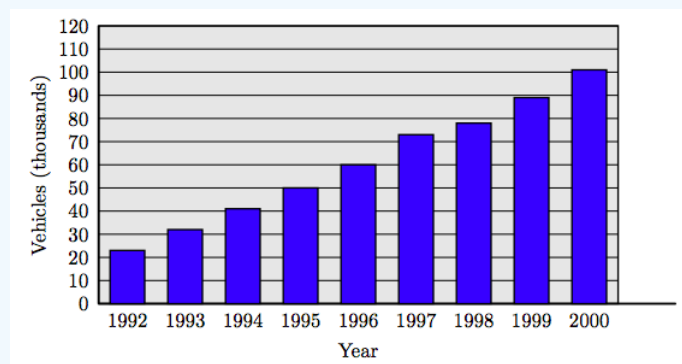
Table 1.2 show the number of cars (in thousands) running on compressed natural gas versus the year. Create a bar chart showing the number of cars running on compressed natural gas versus the year.

Year	1992	1993	1994	1995	1996	1997	1998	1999	2000
Number	23	32	41	50	60	73	78	89	101

Table 1.2: Number of vehicles (in thousands) running on compressed natural gas.

Solution

Place the years on the horizontal axis. At each year, sketch a bar having height equal to the number of cars in that year that are running on compressed natural gas. Scale the vertical axis in thousands.



? Exercise

The following table shows the number of hybrid cars (in thousands) by country.

Country	Number
U.S.	279
Japan	77
Canada	17
U.K.	14
Netherlands	11

Create a bar chart showing the number of cars versus the country of use.

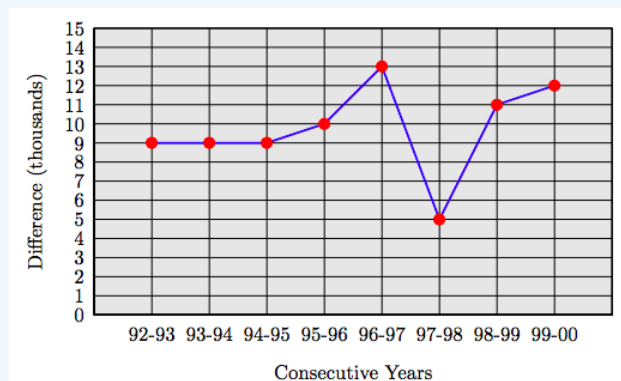
✓ Example 9

Using the data in Table 1.2, create a table that shows the differences in consecutive years, then create a line plot of the result. In what consecutive years did the United States see the greatest increase in cars powered by compressed natural gas?

Solution

Table 1.3 shows the differences in consecutive years.

Next, craft a line graph. Place consecutive years on the horizontal axis. At each consecutive year pair, plot a point at a height equal to the difference in alternative fuel vehicles. Connect the points with straight line segments.



Note how the line graph makes it completely clear that the greatest increase in vehicles powered by compressed natural gas occurred in the consecutive years 1996-1997, an increase of 13,000 vehicles.

? Exercise

The following table show Alphonso's percentage scores on his examinations in mathematics.

Exam	Percentage
Exam #1	52
Exam #2	45
Exam #3	72
Exam #4	88
Exam #5	76

Construct a line graph of Alphonso's exam scores versus exam number.

Exercises

1. Sketch a number line diagram depicting the sum $3 + 2$, as shown in Figure 1.2 in the narrative of this section.
2. Sketch a number line diagram depicting the sum $3 + 5$, as shown in Figure 1.2 in the narrative of this section.
3. Sketch a number line diagram depicting the sum $3 + 4$, as shown in Figure 1.2 in the narrative of this section.
4. Sketch a number line diagram depicting the sum $2 + 4$, as shown in Figure 1.2 in the narrative of this section.
5. Sketch a number line diagram depicting the sum $4 + 2$, as shown in Figure 1.2 in the narrative of this section.
6. Sketch a number line diagram depicting the sum $4 + 3$, as shown in Figure 1.2 in the narrative of this section.
7. Sketch a number line diagram depicting the sum $2 + 5$, as shown in Figure 1.2 in the narrative of this section.
8. Sketch a number line diagram depicting the sum $4 + 5$, as shown in Figure 1.2 in the narrative of this section.
9. Sketch a number line diagram depicting the sum $4 + 4$, as shown in Figure 1.2 in the narrative of this section.
10. Sketch a number line diagram depicting the sum $3 + 3$, as shown in Figure 1.2 in the narrative of this section.

In Exercises 11-28, determine which property of addition is depicted by the given identity

11. $28 + 0 = 28$
12. $53 + 0 = 53$
13. $24 + 0 = 24$
14. $93 + 0 = 93$
15. $(51 + 66) + 88 = 51 + (66 + 88)$
16. $(90 + 96) + 4 = 90 + (96 + 4)$
17. $64 + 39 = 39 + 64$
18. $68 + 73 = 73 + 68$
19. $(70 + 27) + 52 = 70 + (27 + 52)$
20. $(8 + 53) + 81 = 8 + (53 + 81)$
21. $79 + 0 = 79$
22. $42 + 0 = 42$
23. $10 + 94 = 94 + 10$
24. $55 + 86 = 86 + 55$
25. $47 + 26 = 26 + 47$
26. $62 + 26 = 26 + 62$
27. $(61 + 53) + 29 = 61 + (53 + 29)$
28. $(29 + 96) + 61 = 29 + (96 + 61)$

29. Sketch a number line diagram depicting the difference $8 - 2$, as shown in Figure 1.5 in the narrative of this section.
30. Sketch a number line diagram depicting the difference $8 - 4$, as shown in Figure 1.5 in the narrative of this section.
31. Sketch a number line diagram depicting the difference $7 - 2$, as shown in Figure 1.5 in the narrative of this section.
32. Sketch a number line diagram depicting the difference $9 - 5$, as shown in Figure 1.5 in the narrative of this section.
33. Sketch a number line diagram depicting the difference $7 - 4$, as shown in Figure 1.5 in the narrative of this section.
34. Sketch a number line diagram depicting the difference $6 - 4$, as shown in Figure 1.5 in the narrative of this section.

35. Sketch a number line diagram depicting the difference $9-4$, as shown in Figure 1.5 in the narrative of this section.
 36. Sketch a number line diagram depicting the difference $6-5$, as shown in Figure 1.5 in the narrative of this section.
 37. Sketch a number line diagram depicting the difference $8-5$, as shown in Figure 1.5 in the narrative of this section.
 38. Sketch a number line diagram depicting the difference $9-3$, as shown in Figure 1.5 in the narrative of this section.
-

In Exercises 39-50, simplify the given expression.

39. $16 - 8 + 2$
 40. $17 - 3 + 5$
 41. $20 - 5 + 14$
 42. $14 - 5 + 6$
 43. $15 - 2 + 5$
 44. $13 - 4 + 2$
 45. $12 - 5 + 4$
 46. $19 - 4 + 13$
 47. $12 - 6 + 4$
 48. $13 - 4 + 18$
 49. $15 - 5 + 8$
 50. $13 - 3 + 11$
-

In Exercises 51-58, the width W and length L of a rectangle are given. Find the perimeter P of the rectangle.

51. $W = 7$ in, $L = 9$ in
 52. $W = 4$ in, $L = 6$ in
 53. $W = 8$ in, $L = 9$ in
 54. $W = 5$ in, $L = 9$ in
 55. $W = 4$ cm, $L = 6$ cm
 56. $W = 5$ in, $L = 8$ in
 57. $W = 4$ cm, $L = 7$ cm
 58. $W = 4$ in, $L = 9$ in
-

In Exercises 59-66, the length s of a side of a square is given. Find the perimeter P of the square.

59. $s = 25$ cm
 60. $s = 21$ in
 61. $s = 16$ cm
 62. $s = 10$ in
 63. $s = 18$ in
 64. $s = 7$ in
 65. $s = 3$ in
 66. $s = 20$ in
-

In Exercises 67-86, find the sum.

67. $3005 + 5217$

- 68. $1870 + 5021$
 - 69. $575 + 354 + 759$
 - 70. $140 + 962 + 817$
 - 71. $472 + (520 + 575)$
 - 72. $318 + (397 + 437)$
 - 73. $274 + (764 + 690)$
 - 74. $638 + (310 + 447)$
 - 75. $8583 + 592$
 - 76. $5357 + 9936$
 - 77. $899 + 528 + 116$
 - 78. $841 + 368 + 919$
 - 79. $(466 + 744) + 517$
 - 80. $(899 + 996) + 295$
 - 81. $563 + 298 + 611 + 828$
 - 82. $789 + 328 + 887 + 729$
 - 83. $607 + 29 + 270 + 245$
 - 84. $738 + 471 + 876 + 469$
 - 85. $(86 + 557) + 80$
 - 86. $(435 + 124) + 132$
-

In Exercises 87-104, find the difference.

- 87. $3493 - 2034 - 227$
 - 88. $3950 - 1530 - 2363$
 - 89. $8338 - 7366$
 - 90. $2157 - 1224$
 - 91. $2974 - 2374$
 - 92. $881 - 606$
 - 93. $3838 - (777 - 241)$
 - 94. $8695 - (6290 - 4233)$
 - 95. $5846 - 541 - 4577$
 - 96. $5738 - 280 - 4280$
 - 97. $3084 - (2882 - 614)$
 - 98. $1841 - (217 - 28)$
 - 99. $2103 - (1265 - 251)$
 - 100. $1471 - (640 - 50)$
 - 101. $9764 - 4837 - 150$
 - 102. $9626 - 8363 - 1052$
 - 103. $7095 - 226$
 - 104. $4826 - 1199$
-

105. **Water Subsidies.** Since the drought began in 2007, California farms have received \$79 million in water subsidies. California cotton and rice farmers received an additional \$439 million. How much total water subsidies have farmers received? *Associated Press Times-Standard 4/15/09*

106. **War Budget.** The 2010 Federal budget allocates \$534 billion for the Department of Defense base programs and an additional \$130 billion for the nation's two wars. How much will the Department of Defense receive altogether? *Associated Press Times-Standard 5/8/09*

107. **Sun Frost.** Arcata, CA is home to Sun Frost, a manufacturer of highly efficient refrigerators and freezers. The AC model RF12 refrigerator/freezer costs \$2,279 while an R16 model refrigerator/freezer costs \$3,017. How much more does the R16 model cost? Source: www.sunfrost.com/retail/pricelist.html

108. **Shuttle Orbit.** The space shuttle usually orbits at 250 miles above the surface of the earth. To service the Hubble Space Telescope, the shuttle had to go to 350 miles above the surface. How much higher did the shuttle have to orbit?

109. **Earth's Orbit.** Earth orbits the sun in an ellipse. When earth is at its closest to the sun, called *perihelion*, earth is about 147 million kilometers. When earth is at its furthest point from the sun, called *aphelion*, earth is about 152 million kilometers from the sun. What's the difference in millions of kilometers between aphelion and perihelion?

110. **Pluto's Orbit.** Pluto's orbit is highly eccentric. Find the difference between Pluto's closest approach to the sun and Pluto's furthest distance from the sun if Pluto's perihelion (closest point on its orbit about the sun) is about 7 billion kilometers and its aphelion (furthest point on its orbit about the sun) is about 30 billion kilometers.

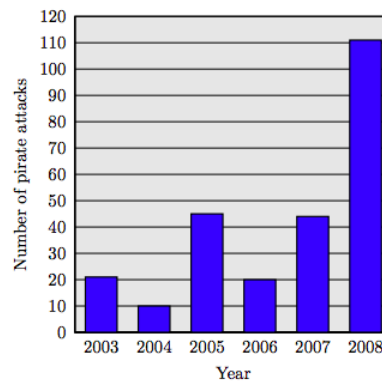
111. **Sunspot Temperature.** The surface of the sun is about 10,000 degrees Fahrenheit. Sunspots are darker regions on the surface of the sun that have a relatively cooler temperature of 6,300 degrees Fahrenheit. How many degrees cooler are sunspots?

112. **Jobs.** The Times-Standard reports that over the next year, the credit- and debitcard processing business Humboldt Merchant Services expects to cut 36 of its 80 jobs, but then turn around and hire another 21. How many people will be working for the company then? *Times-Standard 5/6/09*

113. **Wild tigers.** The chart shows the estimated wild tiger population, by region. According to this chart, what is the total wild tiger population worldwide? *Associated Press-Times-Standard 01/24/10 Pressure mounts to save the tiger.*

Region	Tiger population
India, Nepal and Bhutan	1650
China and Russia	450
Bangladesh	250
Sumatra (Indonesia)	400
Malaysia	500
other SE Asia	350

114. **Pirate Attacks.** The following bar chart tracks pirate attacks off the coast of Somalia.



Source: ICC International Maritime Bureau, AP Times-Standard, 4/15/2009

- How many pirate attacks were there in 2003, 2004, and 2005 combined?
- How many pirate attacks were there in 2006, 2007, and 2008 combined?
- How many more pirate attacks were there in 2008 than in 2007?

115. Emily shows improvement on each successive examination throughout the term. Her exam scores are recorded in the following table.

Exam	Score
Exam #1	48
Exam #2	51
Exam #3	54
Exam #4	59
Exam #5	67
Exam #6	70

- Create a bar plot for Emily's examination scores. Place the examination numbers on the horizontal axis in the same order shown in the table above.
- Create a table that shows successive differences in examination scores. Make a line plot of these differences. Between which two exams did Emily show the greatest improvement?

116. Jason shows improvement on each successive examination throughout the term. His exam scores are recorded in the following table.

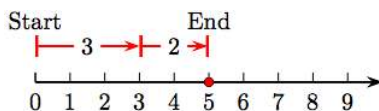
Exam	Score
Exam #1	34
Exam #2	42
Exam #3	45
Exam #4	50
Exam #5	57
Exam #6	62

- Create a bar plot for Jason's examination scores. Place the examination numbers on the horizontal axis in the same order shown in the table above.

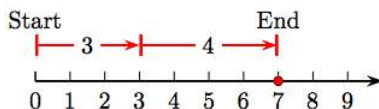
b) Create a table that shows successive differences in examination scores. Make a line plot of these differences.
Between which two exams did Jason show the greatest improvement?

Answers

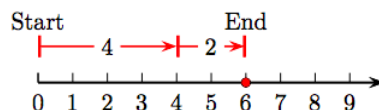
1. $3 + 2 = 5$.



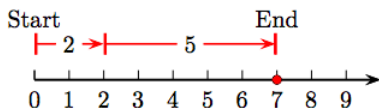
3. $3 + 4 = 7$.



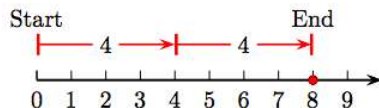
5. $4 + 2 = 6$.



7. $2 + 5 = 7$.



9. $4 + 4 = 8$.



11. Additive identity property of addition.

13. Additive identity property of addition.

15. Associative property of addition

17. Commutative property of addition

19. Associative property of addition

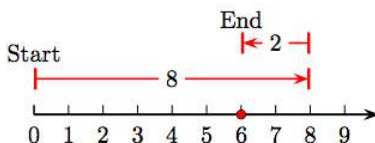
21. Additive identity property of addition.

23. Commutative property of addition

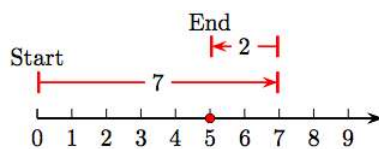
25. Commutative property of addition

27. Associative property of addition

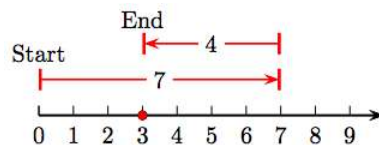
29. $8 - 2 = 6$.



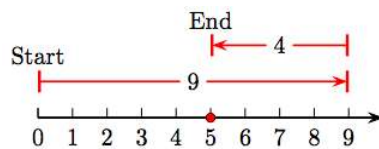
31. $7 - 2 = 5$.



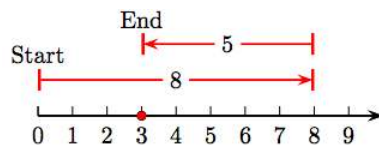
33. $7 - 4 = 3$.



35. $9 - 4 = 5$.



37. $8 - 5 = 3$.



39. 10

41. 29

43. 18

45. 11

47. 10

49. 18

51. $P = 32$ in

53. $P = 34$ in

55. $P = 20$ cm

57. $P = 22$ cm

59. $P = 100$ cm

61. $P = 64$ cm

63. $P = 72$ in

65. $P = 12$ in

67. 8222

69. 1688

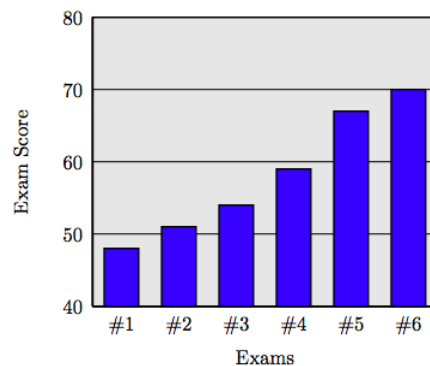
71. 1567

73. 1728

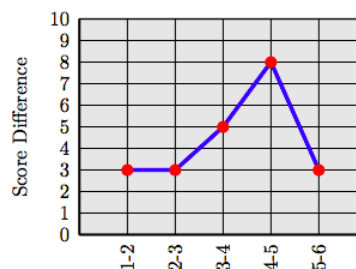
75. 9175

77. 1543

79. 1727
81. 2300
83. 1151
85. 723
87. 1232
89. 972
91. 600
93. 3302
95. 728
97. 816
99. 1089
101. 4777
103. 6869
105. \$518 million
107. \$738
109. 5 million kilometers
111. 3,700 degrees Fahrenheit
113. 3600
115. a) Bar chart.



- b) Line plot of consecutive differences. The line plot of consecutive examination score differences.



The largest improvement was between Exam #4 and Exam #5, where Emily improved by 8 points.

- [1.2: Adding and Subtracting Whole Numbers](#) by [David Arnold](#) is licensed [CC BY-NC-SA 3.0](#).

1.3: Multiplication and Division of Whole Numbers

We begin this section by discussing multiplication of whole numbers. The first order of business is to introduce the various symbols used to indicate multiplication of two whole numbers.

Mathematical symbols that indicate multiplication

Symbol		Example
$\times$	times symbol	3×4
$\cdot$	dot	$3 \cdot 4$
$()$	parentheses	$(3)(4)$ or $3(4)$ or $(3)4$

Products and Factors

In the expression $3 \cdot 4$, the whole numbers 3 and 4 are called the **factors** and $3 \cdot 4$ is called the **product**.

The key to understanding multiplication is held in the following statement.

Multiplication is equivalent to repeated addition.

Suppose, for example, that we would like to evaluate the product $3 \cdot 4$. Because multiplication is equivalent to repeated addition, $3 \cdot 4$ is equivalent to adding three fours. That is,

$$3 \cdot 4 = \underbrace{4 + 4 + 4}_{\text{three fours}}$$

Thus, $3 \cdot 4 = 12$. You can visualize the product $3 \cdot 4$ as the sum of three fours on a number line, as shown in Figure 1.6.

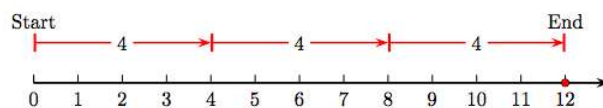


Figure 1.6: Note that $3 \cdot 4 = 4 + 4 + 4$. That is, $3 \cdot 4 = 12$.

Like addition, the order of the factors does not matter.

$$4 \cdot 3 = \underbrace{3 + 3 + 3 + 3}_{\text{four threes}}$$

Thus, $4 \cdot 3 = 12$. Consider the visualization of $4 \cdot 3$ in Figure 1.7.

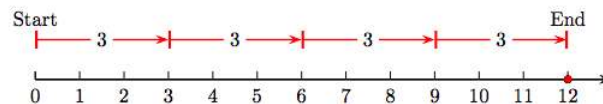


Figure 1.7: Note that $4 \cdot 3 = 3 + 3 + 3 + 3$. That is, $4 \cdot 3 = 12$.

The evidence in Figure 1.6 and Figure 1.7 show us that multiplication is commutative. That is,

$$3 \cdot 4 = 4 \cdot 3$$

Commutative Property of Multiplication

If a and b are any whole numbers, then

$$a \cdot b = b \cdot a.$$

The Multiplicative Identity

In Figure 1.8(a), note that five ones equals 5; that is, $5 \cdot 1 = 5$. On the other hand, in Figure 1.8(b), we see that one five equals five; that is, $1 \cdot 5 = 5$.

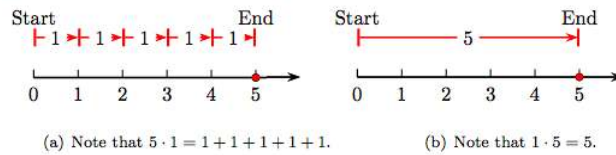


Figure 1.8: Note that $5 \cdot 1 = 5$ and $1 \cdot 5 = 5$.

Because multiplying a whole number by 1 equals that identical number, the whole number 1 is called the multiplicative identity.

The Multiplicative Identity Property

If a is any whole number, then

$$a \cdot 1 = a \text{ and } 1 \cdot a = a.$$

Multiplication by Zero

Because $3 \cdot 4 = 4 + 4 + 4$, we can say that the product $3 \cdot 4$ represents “3 sets of 4,” as depicted in Figure 1.9, where three groups of four boxes are each enveloped in an oval.

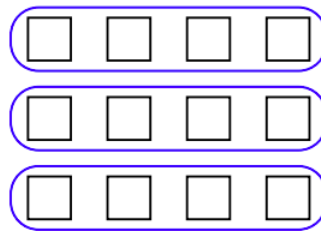


Figure 1.9: Three sets of four: $3 \cdot 4 = 12$.

Therefore, $0 \cdot 4$ would mean zero sets of four. Of course, zero sets of four is zero.

Multiplication by Zero.

If a represents any whole number, then

$$a \cdot 0 = 0 \text{ and } 0 \cdot a = 0.$$

The Associative Property of Multiplication

Like addition, multiplication of whole numbers is associative. Indeed,

$$\begin{aligned} 2 \cdot (3 \cdot 4) &= 2 \cdot 12 \\ &= 24 \end{aligned}$$

and

$$\begin{aligned} (2 \cdot 3) \cdot 4 &= 6 \cdot 4 \\ &= 24. \end{aligned}$$

The Associative Property of Multiplication.

If a , b , and c are any whole numbers, then

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

Multiplying Larger Whole Numbers

Much like addition and subtraction of large whole numbers, we will also need to multiply large whole numbers. Again, we hope the algorithm is familiar from previous coursework.

✓ Example 1

Simplify: $35 \cdot 127$.

Solution

Align the numbers vertically. The order of multiplication does not matter, but we'll put the larger of the two numbers on top of the smaller number. The first step is to multiply 5 times 127. Again, we proceed from right to left. So, 5 times 7 is 35. We write the 5, then carry the 3 to the tens column. Next, 5 times 2 is 10. Add the carry digit 3 to get 13. Write the 3 and carry the 1 to the hundreds column. Finally, 5 times 1 is 5. Add the carry digit to get 6.

$$\begin{array}{r} 1\ 3 \\ 1\ 2\ 7 \\ \times 3\ 5 \\ \hline 6\ 3\ 5 \end{array}$$

The next step is to multiply 3 times 127. However, because 3 is in the tens place, its value is 30, so we actually multiply 30 times 127. This is the same as multiplying 127 by 3 and placing a 0 at the end of the result.

$$\begin{array}{r} 2 \\ 1\ 2\ 7 \\ \times 3\ 5 \\ \hline 6\ 3\ 5 \\ 3\ 8\ 1\ 0 \end{array}$$

After adding the 0, 3 times 7 is 21. We write the 1 and carry the 2 above the 2 in the tens column. Then, 3 times 2 is 6. Add the carry digit 2 to get 8. Finally, 3 times 1 is 3.

All that is left to do is to add the results.

$$\begin{array}{r} 1\ 2\ 7 \\ \times 3\ 5 \\ \hline 6\ 3\ 5 \\ 3\ 8\ 1\ 0 \\ \hline 4\ 4\ 4\ 5 \end{array}$$

Thus, $35 \cdot 127 = 4,445$.

Alternate Format

It does not hurt to omit the trailing zero in the second step of the multiplication, where we multiply 3 times 127. The result would look like this:

$$\begin{array}{r} 1\ 2\ 7 \\ \times 3\ 5 \\ \hline 6\ 3\ 5 \\ 3\ 8\ 1 \\ \hline 4\ 4\ 4\ 5 \end{array}$$

In this format, the zero is understood, so it is not necessary to have it physically present. The idea is that with each multiplication by a new digit, we indent the product one space from the right.

? Exercise

Simplify: $56 \cdot 335$

Answer

18,760

Division of Whole Numbers

We now turn to the topic of division of whole numbers. We first introduce the various symbols used to indicate division of whole numbers.

Mathematical Symbols that Indicate Division

Symbol		Example
$\div$	division symbol	$12 \div 4$
$-$	fraction bar	$\frac{12}{4}$
$\text{\textbackslash longdiv}$	division bar	$4 \text{\textbackslash longdiv} 12$

Note that each of the following say the same thing; that is, “12 divided by 4 is 3.”

$$12 \div 4 = 3 \quad \text{or} \quad \frac{12}{4} = 3 \quad \text{or} \quad 4 \overline{)12}$$

Quotients, Dividends, and Divisors

In the statement

$$\frac{3}{4} \overline{)12}$$

the whole number 12 is called the *dividend*, the whole number 4 is called the *divisor*, and the whole number 3 is called the *quotient*. Note that this division bar notation is equivalent to

$$12 \div 4 = 3 \quad \text{and} \quad \frac{12}{4} = 3.$$

The expression a/b means “ a divided by b ,” but this construct is also called a *fraction*.

Definition: Fraction

The expression

$$\frac{a}{b}$$

is called a **fraction**. The number a on top is called the **numerator** of the fraction; the number b on the bottom is called the **denominator** of the fraction.

The key to understanding division of whole numbers is contained in the following statement.

Division is equivalent to repeated subtraction.

Suppose for example, that we would like to divide the whole number 12 by the whole number 4. This is equivalent to asking the question “how many fours can we subtract from 12?” This can be visualized in a number line diagram, such as the one in Figure 1.10.

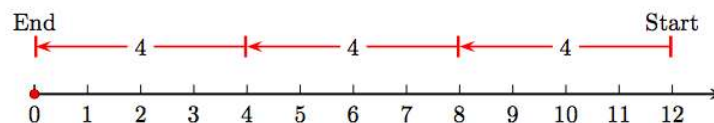


Figure 1.10: Division is repeated subtraction.

In Figure 1.10, note that we if we subtract three fours from twelve, the result is zero. In symbols,

$$12 - \underbrace{4 - 4 - 4}_{\text{three fours}} = 0.$$

Equivalently, we can also ask “How many groups of four are there in 12,” and arrange our work as shown in Figure 1.11, where we can see that in an array of twelve objects, we can circle three groups of four ; i.e., $12 \div 4 = 3$.

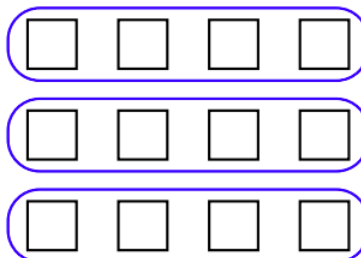


Figure 1.11: There are three groups of four in twelve.

In Figure 1.10 and Figure 1.11, note that the division (repeated subtraction) leaves no remainder. This is not always the case.

✓ Example 2.

Divide 7 by 3.

Solution

In Figure 1.12, we can see that we can subtract two threes from seven, leaving a remainder of one.

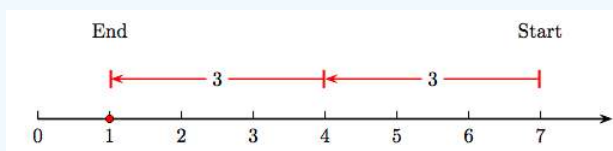


Figure 1.12: Division with a remainder.

Alternatively, in an array of seven objects, we can circle two groups of three, leaving a remainder of one.

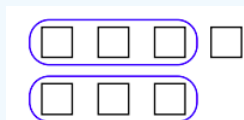


Figure 1.13: Dividing seven by three leaves a remainder of one.

Both Figure 1.12 and Figure 1.13 show that there are two groups of three in seven, with one left over. We say “Seven divided by three is two, with a remainder of one.

? Exercise

Use both the number line approach and the array of boxes approach to divide 12 by 5.

Division is not Commutative

When dividing whole numbers, the order matters. For example

$$12 \div 4 = 3,$$

but $4 \div 12$ is not even a whole number. Thus, if a and b are whole numbers, then $a \div b$ does **not** have to be the same as $b \div a$.

Division is not Associative

When you divide three numbers, the order in which they are grouped will usually affect the answer. For example,

$$\begin{aligned}(48 \div 8) \div 2 &= 6 \div 2 \\ &= 3,\end{aligned}$$

but

$$\begin{aligned}48 \div (8 \div 2) &= 48 \div 4 \\ &= 12.\end{aligned}$$

Thus, if a , b , and c are whole numbers, $(a \div b) \div c$ does not have to be the same as $a \div (b \div c)$.

Division by Zero is Undefined

Suppose that we are asked to divide six by zero; that is, we are asked to calculate $6 \div 0$. In Figure 1.14, we have an array of six objects.

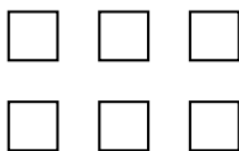


Figure 1.14: How many groups of zero do you see?

Now, to divide six by zero, we must answer the question “How many groups of zero can we circle in Figure 1.14?” Some thought will provide the answer: This is a meaningless request! It makes absolutely no sense to ask how many groups of zero can be circled in the array of six objects in Figure 1.14.

Division by Zero

Division by zero is **undefined**. Each of the expressions

$$6 \div 0 \quad \text{and} \quad \frac{6}{0} \quad \text{and} \quad 0 \overline{)6}$$

is **undefined**.

On the other hand, it make sense to ask “What is zero divided by six?” If we create an array of zero objects, then ask how many groups of six we can circle, the answer is “zero groups of six.” That is, zero divided by six is zero.

$$0 \div 6 = 0 \quad \text{and} \quad \frac{0}{6} = 0 \quad \text{and} \quad 6 \overline{)0}$$

Dividing Larger Whole Numbers

We’ll now provide a quick review of division of larger whole numbers, using an algorithm that is commonly called *long division*. This is not meant to be a thorough discussion, but a cursory one. We’re counting on the fact that our readers have encountered this algorithm in previous courses and are familiar with the process.

✓ Example 3

Simplify: $575/23$.

Solution

We begin by estimating how many times 23 will divide into 57, guessing 1. We put the 1 in the quotient above the 7, multiply 1 times 23, place the answer underneath 57, then subtract.

$$\begin{array}{r} 23 \overline{) 575} \\ \underline{23} \\ 34 \end{array}$$

Because the remainder is larger than the divisor, our estimate is too small. We try again with an estimate of 2.

$$\begin{array}{r} 2 \\ 23 \overline{)575} \\ \underline{46} \\ 115 \\ \underline{115} \\ 0 \end{array}$$

That's the algorithm. Divide, multiply, then subtract. You may continue only when the remainder is smaller than the divisor. To continue, bring down the 5, estimate that 115 divided by 23 is 5, then multiply 5 times the divisor and subtract.

$$\begin{array}{r} 25 \\ 23 \overline{)575} \\ \underline{46} \\ 115 \\ \underline{115} \\ 0 \end{array}$$

Because the remainder is zero, $575/23 = 25$.

? Exercise

Divide $980/35$

Answer

28

Application — Counting Rectangular Arrays

Consider the rectangular array of stars in Figure 1.15. To count the number of stars in the array, we could use brute force, counting each star in the array one at a time, for a total of 20 stars. However, as we have four rows of five stars each, it is much faster to multiply: $4 \cdot 5 = 20$ stars.

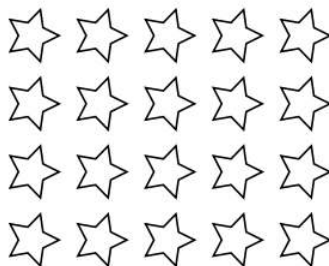


Figure 1.15: Four rows and five columns.

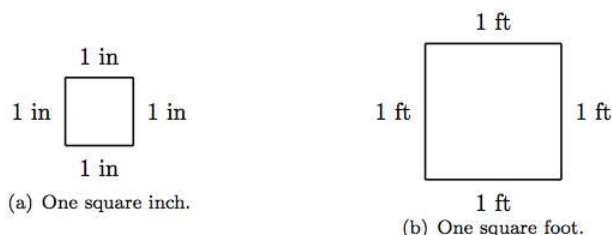


Figure 1.16: Measures of area are in square units.

Application — Area

In Figure 1.16(a), pictured is one square inch (1 in^2), a square with one inch on each side. In Figure 1.16(b), pictured is one square foot (1 ft^2), a square with one foot on each side. Both of these squares are measures of area. Now, consider the rectangle shown in Figure 1.17. The length of this rectangle is four inches (4 in) and the width is three inches (3 in).

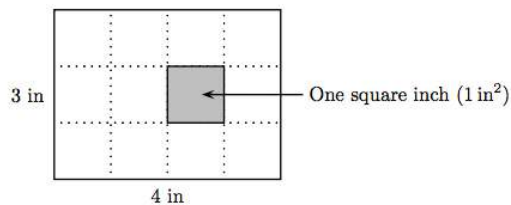


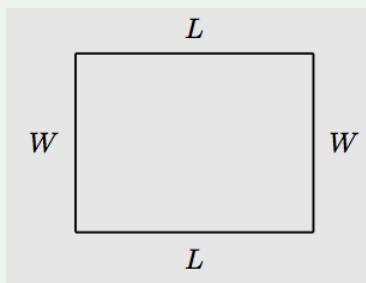
Figure 1.17: A rectangle with length 4 inches and width 3 inches.

To find the area of the figure, we can count the individual units of area that make up the area of the rectangle, twelve square inches (12 in^2) in all. However, as we did in counting the stars in the array in Figure 1.15, it is much faster to note that we have three rows of four square inches. Hence, it is much faster to multiply the number of squares in each row by the number of squares in each column: $4 \cdot 3 = 12$ square inches.

The argument presented above leads to the following rule for finding the area of a rectangle.

Area of a Rectangle

Let L and W represent the length and width of a rectangle, respectively.



To find the area of the rectangle, calculate the product of the length and width. That is, if A represents the area of the rectangle, then the area of the rectangle is given by the formula

$$A = LW.$$

✓ Example 4

A rectangle has width 5 feet and length 12 feet. Find the area of the rectangle.

Solution

Substitute $L = 12 \text{ ft}$ and $W = 5 \text{ ft}$ into the area formula.

$$\begin{aligned} A &= LW \\ &= (12 \text{ ft})(5 \text{ ft}) \\ &= 60 \text{ ft}^2 \end{aligned}$$

Hence, the area of the rectangle is 60 square feet.

? Exercise

A rectangle has width 17 inches and length 33 inches. Find the area of the rectangle.

Answer

561 square inches

Exercises

In Exercises 1-4 use number line diagrams as shown in Figure 1.6 to depict the multiplication.

1. $2 \cdot 4$.

2. $3 \cdot 4$.

3. $4 \cdot 2$.

4. $4 \cdot 3$.

In Exercises 5-16, state the property of multiplication depicted by the given identity.

5. $9 \cdot 8 = 8 \cdot 9$

6. $5 \cdot 8 = 8 \cdot 5$

7. $8 \cdot (5 \cdot 6) = (8 \cdot 5) \cdot 6$

8. $4 \cdot (6 \cdot 5) = (4 \cdot 6) \cdot 5$

9. $6 \cdot 2 = 2 \cdot 6$

10. $8 \cdot 7 = 7 \cdot 8$

11. $3 \cdot (5 \cdot 9) = (3 \cdot 5) \cdot 9$

12. $8 \cdot (6 \cdot 4) = (8 \cdot 6) \cdot 4$

13. $21 \cdot 1 = 21$

14. $39 \cdot 1 = 39$

15. $13 \cdot 1 = 13$

16. $44 \cdot 1 = 44$

In Exercises 17-28, multiply the given numbers.

17. $78 \cdot 3$

18. $58 \cdot 7$

19. $907 \cdot 6$

20. $434 \cdot 80$

21. $128 \cdot 30$

22. $454 \cdot 90$

23. $799 \cdot 60$

24. $907 \cdot 20$

25. $14 \cdot 70$

26. $94 \cdot 90$

27. $34 \cdot 90$

28. $87 \cdot 20$

In Exercises 29-40, multiply the given numbers.

29. $237 \cdot 54$

30. $893 \cdot 94$

31. $691 \cdot 12$

32. $823 \cdot 77$

33. $955 \cdot 89$

34. $714 \cdot 41$

35. $266 \cdot 61$

36. $366 \cdot 31$

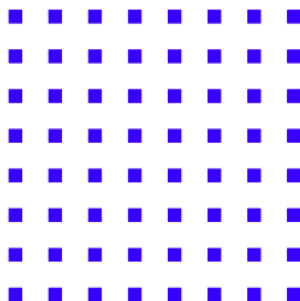
37. $365 \cdot 73$

38. $291 \cdot 47$

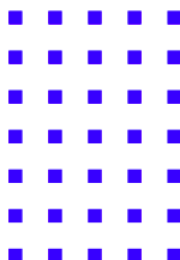
39. $955 \cdot 57$

40. $199 \cdot 33$

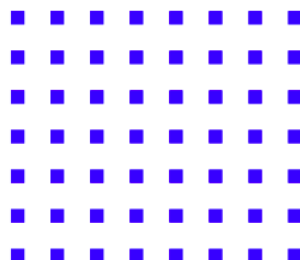
41. Count the number of objects in the array.



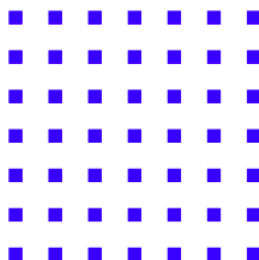
42. Count the number of objects in the array.



43. Count the number of objects in the array.



44. Count the number of objects in the array.



In Exercises 45-48, find the area of the rectangle having the given length and width.

45. $L = 50$ in, $W = 25$ in

46. $L = 48$ in, $W = 24$ in

47. $L = 47$ in, $W = 13$ in

48. $L = 19$ in, $W = 10$ in

In Exercises 49-52, find the perimeter of the rectangle having the given length and width.

49. $L = 25$ in, $W = 16$ in

50. $L = 34$ in, $W = 18$ in

51. $L = 30$ in, $W = 28$ in

52. $L = 41$ in, $W = 25$ in

53. A set of beads costs 50 cents per dozen. What is the cost (in dollars) of 19 dozen sets of beads?

54. A set of beads costs 60 cents per dozen. What is the cost (in dollars) of 7 dozen sets of beads?

55. If a math tutor worked for 47 hours and was paid \$15 each hour, how much money would she have made?

56. If a math tutor worked for 46 hours and was paid \$11 each hour, how much money would he have made?

57. There are 12 eggs in one dozen, and 12 dozen in one gross. How many eggs are in a shipment of 24 gross?

58. There are 12 eggs in one dozen, and 12 dozen in one gross. How many eggs are in a shipment of 11 gross?

59. If bricks weigh 4 kilograms each, what is the weight (in kilograms) of 5000 bricks?

60. If bricks weigh 4 pounds each, what is the weight (in pounds) of 2000 bricks?

In Exercises 61-68, which of the following four expressions differs from the remaining three?

61. $\frac{30}{5}$, $30 \div 5$, $5 \div 30$, $5 \div 30$

62. $\frac{12}{2}$, $12 \div 2$, $2 \div 12$, $2 \div 12$

63. $\frac{8}{2}$, $8 \div 2$, $2 \div 8$, $8 \div 2$

64. $\frac{8}{4}$, $8 \div 4$, $4 \div 8$, $8 \div 4$

65. $2 \div 14$, $14 \div 2$, $\frac{14}{2}$, $14 \div 2$

66. $9 \div 54$, $54 \div 9$, $\frac{54}{9}$, $54 \div 9$

67. $3 \div 24$, $3 \div 24$, $\frac{24}{3}$, $24 \div 3$

68. $3 \div 15$, $3 \div 15$, $\frac{15}{3}$, $15 \div 3$

In Exercises 69-82, simplify the given expression. If the answer doesn't exist or is undefined, write "undefined".

69. $0 \div 11$

70. $0 \div 5$

71. $17 \div 0$

72. $24 \div 0$

73. $10 \cdot 0$

74. $20 \cdot 0$

75. $\frac{7}{0}$

76. $\frac{23}{0}$

77. $16 \div 0$

78. $25 \overline{) 0}$

79. $\frac{0}{24}$

80. $\frac{0}{22}$

81. $0 \overline{) 0}$

82. $0 \div 0$

In Exercises 83-94, divide the given numbers.

83. $\frac{2816}{44}$

84. $\frac{1998}{37}$

85. $\frac{2241}{83}$

86. $\frac{2716}{97}$

87. $\frac{3212}{73}$

88. $\frac{1326}{17}$

89. $\frac{8722}{98}$

90. $\frac{1547}{91}$

91. $\frac{1440}{96}$

92. $\frac{2079}{27}$

93. $\frac{8075}{85}$

94. $\frac{1587}{23}$

In Exercises 95-106, divide the given numbers.

95. $\frac{17756}{92}$

96. $\frac{46904}{82}$

97. $\frac{11951}{19}$

98. $\frac{22304}{41}$

99. $\frac{18048}{32}$

100. $\frac{59986}{89}$

101. $\frac{29047}{31}$

102. $\frac{33264}{86}$

103. $\frac{22578}{53}$

104. $\frac{18952}{46}$

105. $\frac{12894}{14}$

106. $\frac{18830}{35}$

107. A concrete sidewalk is laid in square blocks that measure 6 feet on each side. How many blocks will there be in a walk that is 132 feet long?

108. A concrete sidewalk is laid in square blocks that measure 5 feet on each side. How many blocks will there be in a walk that is 180 feet long?

109. One boat to the island can take 5 people. How many trips will the boat have to take in order to ferry 38 people to the island? (Hint: Round up your answer.)

110. One boat to the island can take 4 people. How many trips will the boat have to take in order to ferry 46 people to the island? (Hint: Round up your answer.)
111. If street lights are placed at most 145 feet apart, how many street lights will be needed for a street that is 4 miles long, assuming that there are lights at each end of the street? (Note: 1 mile = 5280 feet.)
112. If street lights are placed at most 70 feet apart, how many street lights will be needed for a street that is 3 miles long, assuming that there are lights at each end of the street? (Note: 1 mile = 5280 feet.)
113. A concrete sidewalk is laid in square blocks that measure 4 feet on each side. How many blocks will there be in a walk that is 292 feet long?
114. A concrete sidewalk is laid in square blocks that measure 5 feet on each side. How many blocks will there be in a walk that is 445 feet long?
115. One boat to the island can take 3 people. How many trips will the boat have to take in order to ferry 32 people to the island? (Hint: Round up your answer.)
116. One boat to the island can take 4 people. How many trips will the boat have to take in order to ferry 37 people to the island? (Hint: Round up your answer.)
117. If street lights are placed at most 105 feet apart, how many street lights will be needed for a street that is 2 miles long, assuming that there are lights at each end of the street? (Note: 1 mile = 5280 feet.)
118. If street lights are placed at most 105 feet apart, how many street lights will be needed for a street that is 3 miles long, assuming that there are lights at each end of the street? (Note: 1 mile = 5280 feet.)
-
119. **Writing articles.** Eli writes an average of 4 articles a day, five days a week, to support product sales. How many articles does Eli write in one week?
120. **Machine gun.** A 0.50-caliber antiaircraft machine gun can fire 800 rounds each minute. How many rounds could fire in three minutes? *Associated Press Times-Standard 4/15/09*
121. **Laps.** The swimming pool at CalCourts is 25 yards long. If one lap is up and back again, how many yards has Wendell swam doing 27 laps?
122. **Refrigerator wattage.** A conventional refrigerator will run about 12 hours each day can use 150 Watts of power each hour. How many Watts of power will a refrigerator use over the day?
123. **Horse hay.** A full-grown horse should eat a minimum of 12 pounds of hay each day and may eat much more depending on their weight. How many pounds minimum would a horse eat over a year?
124. **College costs.** After a \$662 hike in fees, California residents who want to attend the University of California as an undergraduate should expect to pay \$8,700 in for the upcoming academic year 2009- 2010. If the cost were to remain the same for the next several years, how much should a student expect to pay for a four-year degree program at a UC school?
125. **Non-resident costs.** Nonresident undergraduates who want to attend a University of California college should expect to pay about \$22,000 for the upcoming academic year. Assuming costs remain the same, what can a four-year degree cost?
126. **Student tax.** The mayer of Providence, Rhode Island wants to tax its 25,000 Brown University students \$150 each to contribute to tax receipts saying students should pay for the resources they use just like the town residents. How many dollars would the mayer generate?
127. **New iceberg.** A new iceberg, shaved off a glacier after a collision with another iceberg, measures about 48 miles long and 28 miles wide. What's the approximate area of the new iceberg? *Associated PressTimes-Standard 02/27/10* 2 Huge icebergs set loose off Antarctica's coast.
128. **Solar panels.** One of the solar panels on the International Space Station is 34 meters long and 11 meters wide. If there are eight of these, what's the total area for solar collection?
129. **Sidewalk.** A concrete sidewalk is to be 80 foot long and 4 foot wide. How much will it cost to lay the sidewalk at \$8 per square foot?

130. **Hay bales.** An average bale of hay weighs about 60 pounds. If a horse eats 12 pounds of hay a day, how many days will one bale feed a horse?

131. **Sunspots.** Sunspots, where the sun's magnetic field is much higher, usually occur in pairs. If the total count of sunspots is 72, how many pairs of sunspots are there?

Answers

1. $2 \cdot 4 = \underbrace{4 + 4}_{2 \text{ times}} = 8$

3. $4 \cdot 2 = \underbrace{2 + 2 + 2 + 2}_{4 \text{ times}} = 8$

5. Commutative property of multiplication

7. Associative property of multiplication

9. Commutative property of multiplication

11. Associative property of multiplication

13. Multiplicative identity property

15. Multiplicative identity property

17. 234

19. 5442

21. 3840

23. 47940

25. 980

27. 3060

29. 12798

31. 8292

33. 84995

35. 16226

37. 26645

39. 54435

41. 64

43. 56

45. 1250 in^2

47. 611 in^2

49. 82 in

51. 116 in

53. 9.50

55. 705

57. 3456

59. 20000

61. $5 \div 30$

63. $8\sqrt{2}$

65. $14\sqrt{2}$

67. $3 \div 24$

69. 0

71. Undefined

73. 0

75. Undefined

77. 0

79. 0

81. Undefined

83. 64

85. 27

87. 44

89. 89

91. 15

93. 95

95. 193

97. 629

99. 564

101. 937

103. 426

105. 921

107. 22

109. 8

111. 147

113. 73

115. 11

117. 102

119. 20 articles

121. 1350 yards

123. 4380 pounds of hay

125. \$88,000

127. 1344 mi^2

129. \$2,560

131. 36

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1.4: Prime Factorization

In the statement $3 \cdot 4 = 12$, the number 12 is called the *product*, while 3 and 4 are called *factors*.

✓ Example 1

Find all whole number factors of 18.

Solution

We need to find all whole number pairs whose product equals 18. The following pairs come to mind.

$$1 \cdot 18 = 18 \text{ and } 2 \cdot 9 = 18 \text{ and } 3 \cdot 6 = 18.$$

Hence, the factors of 18 are (in order) 1, 2, 3, 6, 9, and 18.

? Exercise

Find all whole number factors of 21.

Answer

1, 3, 7, and 21

Divisibility

In Example 1, we saw $3 \cdot 6 = 18$, making 3 and 6 factors of 18. Because division is the inverse of multiplication, that is, division by a number undoes the multiplication of that number, this immediately provides

$$18 \div 6 = 3 \text{ and } 18 \div 3 = 6.$$

That is, 18 is divisible by 3 and 18 is divisible by 6. When we say that 18 is divisible by 3, we mean that when 18 is divided by 3, there is a zero remainder.

Divisible

Let a and b be whole numbers. Then a is divisible by b if and only if the remainder is zero when a is divided by b . In this case, we say that “ b is a divisor of a .”

✓ Example 2

Find all whole number divisors of 18.

Solution

In Example 1, we saw that $3 \cdot 6 = 18$. Therefore, 18 is divisible by both 3 and 6 ($18 \div 3 = 6$ and $18 \div 6 = 3$). Hence, when 18 is divided by 3 or 6, the remainder is zero. Therefore, 3 and 6 are divisors of 18. Noting the other products in Example 1, the complete list of divisors of 18 is 1, 2, 3, 6, 9, and 18.

? Exercise

Find all whole number divisors of 21.

Answer

1, 3, 7, and 21.

Example 1 and Example 2 show that when working with whole numbers, the words *factor* and *divisor* are interchangeable.

Factors and Divisors

If $c = a \cdot b$, then a and b are called *factors* of c . Both a and b are also called *divisors* of c .

Divisibility Tests

There are a number of very useful divisibility tests.

Divisible by 2. If a whole number ends in 0, 2, 4, 6, or 8, then the number is called an even number and is divisible by 2. Examples of even numbers are 238 and 1,246 ($238 \div 2 = 119$ and $1,246 \div 2 = 623$). A number that is not even is called an odd number. Examples of odd numbers are 113 and 2,339.

Divisible by 3. If the sum of the digits of a whole number is divisible by 3, then the number itself is divisible by 3. An example is 141. The sum of the digits is $1 + 4 + 1 = 6$, which is divisible by 3. Therefore, 141 is also divisible by 3 ($141 \div 3 = 47$).

Divisible by 4. If the number represented by the last two digits of a whole number is divisible by 4, then the number itself is divisible by 4. An example is 11,524. The last two digits represent 24, which is divisible by 4 ($24 \div 4 = 6$). Therefore, 11,524 is divisible by 4 ($11,524 \div 4 = 2,881$).

Divisible by 5. If a whole number ends in a zero or a 5, then the number is divisible by 5. Examples are 715 and 120 ($715 \div 5 = 143$ and $120 \div 5 = 24$).

Divisible by 6. If a whole number is divisible by 2 and by 3, then it is divisible by 6. An example is 738. First, 738 is even and divisible by 2. Second, $7+3+8=18$, which is divisible by 3. Hence, 738 is divisible by 3. Because 738 is divisible by both 2 and 3, it is divisible by 6 ($738 \div 6 = 123$).

Divisible by 8. If the number represented by the last three digits of a whole number is divisible by 8, then the number itself is divisible by 8. An example is 73,024. The last three digits represent the number 024, which is divisible by 8 ($024 \div 8 = 3$). Thus, 73,024 is also divisible by 8 ($73,024 \div 8 = 9,128$).

Divisible by 9. If the sum of the digits of a whole number is divisible by 9, then the number itself is divisible by 9. An example is 117. The sum of the digits is $1 + 1 + 7 = 9$, which is divisible by 9. Hence, 117 is divisible by 9 ($117 \div 9 = 13$).

Prime Numbers

We begin with the definition of a prime number.

Prime Number

A whole number (other than 1) is a prime number if its only factors (divisors) are 1 and itself. Equivalently, a number is prime if and only if it has exactly two factors (divisors).

✓ Example 3

Which of the whole numbers 12, 13, 21, and 37 are prime numbers?

Solution

- The factors (divisors) of 12 are 1, 2, 3, 4, 6, and 12. Hence, 12 is not a prime number.
- The factors (divisors) of 13 are 1 and 13. Because its only divisors are 1 and itself, 13 is a prime number.
- The factors (divisors) of 21 are 1, 3, 7, and 21. Hence, 21 is not a prime number.
- The factors (divisors) of 37 are 1 and 37. Because its only divisors are 1 and itself, 37 is a prime number.

? Exercise

Which of the whole numbers 15, 23, 51, and 59 are prime numbers?

Answer

23 and 59

✓ Example 4

List all the prime numbers less than 20.

Solution

The prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, and 19.

You try it!

List all the prime numbers less than 100.

Composite Numbers

If a whole number is not a prime number, then it is called a *composite number*.

✓ Example 5

Is the whole number 1,179 prime or composite?

Solution

Note that $1 + 1 + 7 + 9 = 18$, which is divisible by both 3 and 9. Hence, 3 and 9 are both divisors of 1,179. Therefore, 1,179 is a composite number.

? Exercise

Is the whole number 2,571 prime or composite?

Answer

Composite

Factor Trees

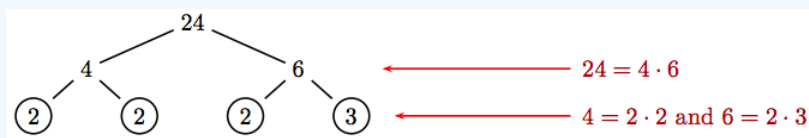
We will now learn how to express a composite number as a unique product of prime numbers. The most popular device for accomplishing this goal is the *factor tree*.

✓ Example 6

Express 24 as a product of prime factors.

Solution

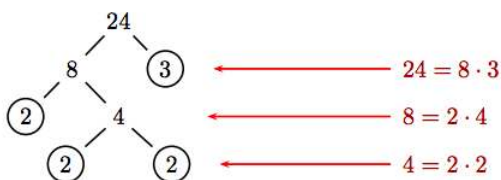
We use a factor tree to break 24 down into a product of primes.



At each level of the tree, break the current number into a product of two factors. The process is complete when all of the “circled leaves” at the bottom of the tree are prime numbers. Arranging the factors in the “circled leaves” in order,

$$24 = 2 \cdot 2 \cdot 2 \cdot 3.$$

The final answer does not depend on product choices made at each level of the tree. Here is another approach.



The final answer is found by including all of the factors from the “circled leaves” at the end of each branch of the tree, which yields the same result, namely $24 = 2 \cdot 2 \cdot 2 \cdot 3$.

Alternate Approach

Some favor repeatedly dividing by 2 until the result is no longer divisible by 2. Then try repeatedly dividing by the next prime until the result is no longer divisible by that prime. The process terminates when the last resulting quotient is equal to the number 1.

2	24	
2	12	$24 \div 2 = 12$
2	6	$12 \div 2 = 6$
3	3	$6 \div 2 = 3$
	1	$3 \div 3 = 1$

The first column reveals the prime factorization; i.e., $24 = 2 \cdot 2 \cdot 2 \cdot 3$.

? Exercise

Express 36 as a product of prime factors.

Answer

$2 \cdot 2 \cdot 3 \cdot 3$.

The fact that the alternate approach in Example 6 yielded the same result is significant.

Unique Factorization Theorem

Every whole number can be **uniquely** factored as a product of primes.

This result guarantees that if the prime factors are ordered from smallest to largest, everyone will get the same result when breaking a number into a product of prime factors.

Exponents

We begin with the definition of an exponential expression.

Exponents

The expression a^m is defined to mean

$$a^m = \underbrace{a \cdot a \cdot \dots \cdot a}_{m \text{ times}}$$

The number a is called the base of the exponential expression and the number m is called the exponent. The exponent m tells us to repeat the base a as a factor m times.

✓ Example 7

Evaluate 2^5 , 2^3 , and 5^2 .

Solution

- In the case of 2^5 , we have

$$\begin{aligned}2^5 &= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \\ &= 32.\end{aligned}$$

- In the case of 3^3 , we have

$$\begin{aligned}3^3 &= 3 \cdot 3 \cdot 3 \\ &= 27.\end{aligned}$$

- In the case of 5^2 , we have

$$\begin{aligned}5^2 &= 5 \cdot 5 \\ &= 25.\end{aligned}$$

? Exercise

Evaluate: 3^5 .

Answer

243.

✓ Example 8

Express the solution to Example 6 in compact form using exponents.

Solution

In Example 6, we determined the prime factorization of 24.

$$24 = 2 \cdot 2 \cdot 2 \cdot 3$$

Because $2 \cdot 2 \cdot 2 = 8$, we can write this more compactly.

$$24 = 2^3 \cdot 3$$

? Exercise

Prime factor 54.

Answer

$$2 \cdot 3 \cdot 3 \cdot 3$$

✓ Example 9

Evaluate the expression $2^3 \cdot 3^2 \cdot 5^2$.

Solution

First raise each factor to the given exponent, then perform the multiplication in order (left to right).

$$\begin{aligned}2^3 \cdot 3^2 \cdot 5^2 &= 8 \cdot 9 \cdot 25 \\ &= 72 \cdot 25 \\ &= 1800\end{aligned}$$

? Exercise

Evaluate: $3^3 \cdot 5^2$.

Answer

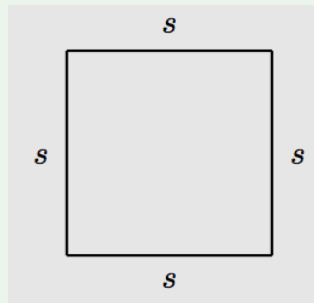
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Application

A square is a rectangle with four equal sides.

Area of a Square

Let s represent the length of each side of a square.



Because a square is also a rectangle, we can find the area of the square by multiplying its length and width. However, in this case, the length and width both equal s , so $A = (s)(s) = s^2$. Hence, the formula for the area of a square is

$$A = s^2.$$

✓ Example 10

The edge of a square is 13 centimeters. Find the area of a square.

Solution

Substitute $s = 13$ cm into the area formula.

$$\begin{aligned} A &= s^2 \\ &= (13 \text{ cm})^2 \\ &= (13 \text{ cm})(13 \text{ cm}) \\ &= 169 \text{ cm}^2 \end{aligned}$$

Hence, the area of the square is 169 cm^2 ; i.e., 169 square centimeters.

? Exercise

The edge of a square is 15 meters. Find the area of a square.

Answer

225 square meters

Exercises

In Exercises 1-12, find all divisors of the given number.

1. 30

2. 19
3. 83
4. 51
5. 91
6. 49
7. 75
8. 67
9. 64
10. 87
11. 14
12. 89

In Exercises 13-20, which of the following numbers is not divisible by 2?

13. 117, 120, 342, 230
14. 310, 157, 462, 160
15. 30, 22, 16, 13
16. 382, 570, 193, 196
17. 105, 206, 108, 306
18. 60, 26, 23, 42
19. 84, 34, 31, 58
20. 66, 122, 180, 63

In Exercises 21-28, which of the following numbers is not divisible by 3?

21. 561, 364, 846, 564
22. 711, 850, 633, 717
23. 186, 804, 315, 550
24. 783, 909, 504, 895
25. 789, 820, 414, 663
26. 325, 501, 945, 381
27. 600, 150, 330, 493
28. 396, 181, 351, 606

In Exercises 29-36, which of the following numbers is not divisible by 4?

29. 3797, 7648, 9944, 4048
30. 1012, 9928, 7177, 1592
31. 9336, 9701, 4184, 2460
32. 2716, 1685, 2260, 9788
33. 9816, 7517, 8332, 7408
34. 1788, 8157, 7368, 4900
35. 1916, 1244, 7312, 7033

36. 7740, 5844, 2545, 9368

In Exercises 37-44, which of the following numbers is not divisible by 5?

37. 8920, 4120, 5285, 9896

38. 3525, 7040, 2185, 2442

39. 8758, 3005, 8915, 3695

40. 3340, 1540, 2485, 2543

41. 2363, 5235, 4145, 4240

42. 9030, 8000, 5445, 1238

43. 1269, 5550, 4065, 5165

44. 7871, 9595, 3745, 4480

In Exercises 45-52, which of the following numbers is not divisible by 6?

45. 328, 372, 990, 528

46. 720, 288, 148, 966

47. 744, 174, 924, 538

48. 858, 964, 930, 330

49. 586, 234, 636, 474

50. 618, 372, 262, 558

51. 702, 168, 678, 658

52. 780, 336, 742, 312

In Exercises 53-60, which of the following numbers is not divisible by 8?

53. 1792, 8216, 2640, 5418

54. 2168, 2826, 1104, 2816

55. 8506, 3208, 9016, 2208

56. 2626, 5016, 1392, 1736

57. 4712, 3192, 2594, 7640

58. 9050, 9808, 8408, 7280

59. 9808, 1232, 7850, 7912

60. 3312, 1736, 9338, 3912

In Exercises 61-68, which of the following numbers is not divisible by 9?

61. 477, 297, 216, 991

62. 153, 981, 909, 919

63. 153, 234, 937, 675

64. 343, 756, 927, 891

65. 216, 783, 594, 928

66. 504, 279, 307, 432

67. 423, 801, 676, 936

68. 396, 684, 567, 388

In Exercises 69-80, identify the given number as prime, composite, or neither.

69. 19

70. 95

71. 41

72. 88

73. 27

74. 61

75. 91

76. 72

77. 21

78. 65

79. 23

80. 36

In Exercises 81-98, find the prime factorization of the natural number.

81. 224

82. 320

83. 108

84. 96

85. 243

86. 324

87. 160

88. 252

89. 32

90. 128

91. 360

92. 72

93. 144

94. 64

95. 48

96. 200

97. 216

98. 392

In Exercises 99-110, compute the exact value of the given exponential expression.

99. $5^2 \cdot 4^1$

100. $2^3 \cdot 4^1$

101. 0^1

102. 1^3

103. $3^3 \cdot 0^2$

104. $3^3 \cdot 2^2$

105. 4^1

106. 5^2

107. 4^3

108. 4^2

109. $3^3 \cdot 1^2$

110. $5^2 \cdot 2^3$

In Exercises 111-114, find the area of the square with the given side.

111. 28 inches

112. 31 inches

113. 22 inches

114. 13 inches

Create factor trees for each number in Exercises 115-122. Write the prime factorization for each number in compact form, using exponents.

115. 12

116. 18

117. 105

118. 70

119. 56

120. 56

121. 72

122. 270

123. **Sieve of Eratosthenes.** This exercise introduces the *Sieve of Eratosthenes*, an ancient algorithm for finding the primes less than a certain number n , first created by the Greek mathematician Eratosthenes. Consider the grid of integers from 2 through 100.

2	3	4	5	6	7	8	9	10	11
12	13	14	15	16	17	18	19	20	21
22	23	24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39	40	41
42	43	44	45	46	47	48	49	50	51
52	53	54	55	56	57	58	59	60	61
62	63	64	65	66	67	68	69	70	71
72	73	74	75	76	77	78	79	80	81
82	83	84	85	86	87	88	89	90	91
92	93	94	95	96	97	98	99	100	

To find the primes less than 100, proceed as follows.

- i) Strike out all multiples of 2 (4, 6, 8, etc.)
- ii) The list's next number that has not been struck out is a prime number.
- iii) Strike out from the list all multiples of the number you identified in step (ii).
- iv) Repeat steps (ii) and (iii) until you can no longer strike any more multiples.
- v) All unstruck numbers in the list are primes.

Answers

- 1. 1, 2, 3, 5, 6, 10, 15, 30
- 3. 1, 83
- 5. 1, 7, 13, 91
- 7. 1, 3, 5, 15, 25, 75
- 9. 1, 2, 4, 8, 16, 32, 64
- 11. 1, 2, 7, 14
- 13. 117
- 15. 13
- 17. 105
- 19. 31
- 21. 364
- 23. 550
- 25. 820
- 27. 493
- 29. 3797
- 31. 9701
- 33. 7517
- 35. 7033
- 37. 9896
- 39. 8758
- 41. 2363
- 43. 1269
- 45. 328
- 47. 538
- 49. 586
- 51. 658
- 53. 5418
- 55. 8506
- 57. 2594
- 59. 7850
- 61. 991
- 63. 937

65. 928

67. 676

69. prime

71. prime

73. composite

75. composite

77. composite

79. prime

81. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 7$

83. $2 \cdot 2 \cdot 3 \cdot 3 \cdot 3$

85. $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$

87. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5$

89. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

91. $2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 5$

93. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3$

95. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3$

97. $2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3$

99. 100

101. 0

103. 0

105. 4

107. 64

109. 27

111. 784 in^2

113. 484 in^2

115. $12 = 2 \cdot 2 \cdot 3$

117. $105 = 3 \cdot 5 \cdot 7$

119. $56 = 2^3 \cdot 7$

121. $72 = 2^3 \cdot 3^2$

123. Unstruck numbers are primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

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1.5: Order of Operations

The order in which we evaluate expressions can be ambiguous. Take for example, the expression $4 + 3 \cdot 2$. If we do the addition first, then

$$\begin{aligned} 4 + 3 \cdot 2 &= 7 \cdot 2 \\ &= 14. \end{aligned}$$

On the other hand, if we do the multiplication first, then

$$\begin{aligned} 4 + 3 \cdot 2 &= 4 + 6 \\ &= 10. \end{aligned}$$

So, what are we to do? Of course, grouping symbols can remove the ambiguity

Grouping Symbols

Parentheses, brackets, or curly braces can be used to group parts of an expression. Each of the following are equivalent:

$$(4 + 3) \cdot 2 \text{ or } [4 + 3] \cdot 2 \text{ or } \{4 + 3\} \cdot 2$$

In each case, the rule is “evaluate the expression inside the grouping symbols first.” If grouping symbols are nested, evaluate the expression in the innermost pair of grouping symbols first.

Thus, for example,

$$\begin{aligned} (4 + 3) \cdot 2 &= 7 \cdot 2 \\ &= 14. \end{aligned}$$

Note how the expression contained in the parentheses was evaluated first. Another way to avoid ambiguities in evaluating expressions is to establish an order in which operations should be performed. The following guidelines should always be strictly enforced when evaluating expressions.

Rules Guiding Order of Operations

When evaluating expressions, proceed in the following order.

1. Evaluate expressions contained in grouping symbols first. If grouping symbols are nested, evaluate the expression in the innermost pair of grouping symbols first.
2. Evaluate all exponents that appear in the expression.
3. Perform all multiplications and divisions in the order that they appear in the expression, moving left to right.
4. Perform all additions and subtractions in

✓ Example 1

Evaluate $4 + 3 \cdot 2$.

Solution

Because of the established *Rules Guiding Order of Operations*, this expression is no longer ambiguous. There are no grouping symbols or exponents, so we immediately go to rule three, evaluate all multiplications and divisions in the order that they appear, moving left to right. After that we invoke rule four, performing all additions and subtractions in the order that they appear, moving left to right.

$$\begin{aligned} 4 + 3 \cdot 2 &= 4 + 6 \\ &= 10 \end{aligned}$$

Thus, $4 + 3 \cdot 2 = 10$.

? Exercise

Simplify: $8 + 2 \cdot 5$.

Answer

18

✓ Example 2

Evaluate $18 - 2 + 3$.

Solution

Follow the *Rules Guiding Order of Operations*. Addition has no precedence over subtraction, nor does subtraction have precedence over addition. We are to perform additions and subtractions as they occur, moving left to right.

$$\begin{aligned} 18 - 2 + 3 &= 16 + 3 \text{ Subtract: } 18 - 2 = 16. \\ &= 19 \text{ Add: } 16 + 3 = 19. \end{aligned}$$

Thus, $18 - 2 + 3 = 19$.

? Exercise

Simplify: $17 - 8 + 2$.

Answer

11

✓ Example 3

Evaluate $54 \div 9 \cdot 2$.

Solution

Follow the *Rules Guiding Order of Operations*. Division has no precedence over multiplication, nor does multiplication have precedence over division. We are to perform divisions and multiplications as they occur, moving left to right.

$$\begin{aligned} 54 \div 9 \cdot 2 &= 6 \cdot 2 \text{ Divide: } 54 \div 9 = 6. \\ &= 12 \text{ Multiply: } 6 \cdot 2 = 12. \end{aligned}$$

Thus, $54 \div 9 \cdot 2 = 12$.

? Exercise

Simplify: $72 \div 9 \cdot 2$.

Answer

16

✓ Example 4

Evaluate $2 \cdot 3^2 - 12$.

Solution

Follow the *Rules Guiding Order of Operations*, exponents first, then multiplication, then subtraction.

 allowed only in math mode

Thus, $2 \cdot 3^2 - 12 = 6$.

? Exercise

Simplify: $14 + 3 \cdot 4^2$

Answer

62

✓ Example 5

Evaluate $12 + 2(3 + 2 \cdot 5)^2$.

Solution

Follow the Rules Guiding Order of Operations, evaluate the expression inside the parentheses first, then exponents, then multiplication, then addition.

$$\begin{aligned} 12 + 2(3 + 2 \cdot 5)^2 &= 12 + 2(3 + 10)^2 && \text{Multiply inside parentheses: } 2 \cdot 5 = 10. \\ &= 12 + 2(13)^2 && \text{Add inside parentheses: } 3 + 10 = 13. \\ &= 12 + 2(169) && \text{Exponents are next: } (13)^2 = 169. \\ &= 12 + 338 && \text{Multiplication is next: } 2(169) = 338. \\ &= 350 && \text{Time to add: } 12 + 338 = 350. \end{aligned}$$

Thus, $12 + 2(3 + 2 \cdot 5)^2 = 350$.

? Exercise

Simplify: $3(2 + 3 \cdot 4)^2 - 11$.

Answer

577

✓ Example 6

Evaluate $2\{2 + 2[2 + 2]\}$.

Solution

When grouping symbols are nested, evaluate the expression between the pair of innermost grouping symbols first.

$$\begin{aligned} 2(2 + 2[2 + 2]) &= 2(2 + 2[4]) && \text{Innermost grouping first: } 2 + 2 = 4. \\ &= 2(2 + 8) && \text{Multiply next: } 2[4] = 8. \\ &= 2(10) && \text{Add inside braces: } 2 + 8 = 10. \\ &= 20 && \text{Multiply: } 2(10) = 20 \end{aligned}$$

Thus, $2(2 + 2[2 + 2]) = 20$.

? Exercise

Simplify: $2\{3 + 2[3 + 2]\}$.

Answer

26

Fraction Bars

Consider the expression

$$\frac{6^2 + 8^2}{(2 + 3)^2}$$

Because a fraction bar means division, the above expression is equivalent to

$$(6^2 + 8^2) \div (2 + 3)^2$$

The position of the grouping symbols signals how we should proceed. We should simplify the numerator, then the denominator, then divide.

Fractional Expressions

If a fractional expression is present, evaluate the numerator and denominator first, then divide.

✓ Example 7

Evaluate the expression

$$\frac{6^2 + 8^2}{(2 + 3)^2}$$

Solution

Simplify the numerator and denominator first, then divide.

$$\begin{aligned} \frac{6^2 + 8^2}{(2 + 3)^2} &= \frac{6^2 + 8^2}{(5)^2} && \text{Parentheses in denominator first: } 2 + 3 = 5 \\ &= \frac{36 + 64}{25} && \text{Exponents are next: } 6^2 = 36, 8^2 = 64, 5^2 = 25. \\ &= \frac{100}{25} && \text{Add in numerator: } 36 + 64 = 100 \\ &= 4 && \text{Divide: } 100 \div 25 = 4. \end{aligned}$$

Thus, $\frac{6^2 + 8^2}{(2 + 3)^2} = 4$.

? Exercise

Simplify: $\frac{12 + 3 \cdot 2}{6}$ **Answer**

3

The Distributive Property

Consider the expression $2 \cdot (3 + 4)$. If we follow the “Rules Guiding Order of Operations,” we would evaluate the expression inside the parentheses first. $2 \cdot (3 + 4) = 2 \cdot 7$ Parentheses first: $3 + 4 = 7$. $= 14$ Multiply: $2 \cdot 7 = 14$.

However, we could also choose to “distribute” the 2, first multiplying 2 times each addend in the parentheses.

$$\begin{aligned} 2 \cdot (3 + 4) &= 2 \cdot 3 + 2 \cdot 4 && \text{Multiply 2 times both 3 and 4.} \\ &= 6 + 8 && \text{Multiply: } 2 \cdot 3 = 6 \text{ and } 2 \cdot 4 = 8. \\ &= 14 && \text{Add: } 6 + 8 = 14. \end{aligned}$$

The fact that we get the same answer in the second approach is an illustration of an important property of whole numbers.¹

The Distributive Property

Let a , b , and c be any whole numbers. Then,

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

We say that “multiplication is distributive with respect to addition.”

Multiplication is distributive with respect to addition. If you are not computing the product of a number and a sum of numbers, the distributive property does not apply.

Caution! Wrong Answer Ahead!

If you are calculating the product of a number and the product of two numbers, the distributive property must not be used. For example, here is a common misapplication of the distributive property.

$$\begin{aligned} 2 \cdot (3 \cdot 4) &= (2 \cdot 3) \cdot (2 \cdot 4) \\ &= 6 \cdot 8 \\ &= 48 \end{aligned}$$

This result is quite distant from the correct answer, which is found by computing the product within the parentheses first.

$$\begin{aligned} 2 \cdot (3 \cdot 4) &= 2 \cdot 12 \\ &= 24. \end{aligned}$$

In order to apply the distributive property, you must be multiplying times a sum.

Example 8

Use the distributive property to calculate $4 \cdot (5 + 11)$.

Solution

This is the product of a number and a sum, so the distributive property may be applied.

$$\begin{aligned} 4 \cdot (5 + 11) &= 4 \cdot 5 + 4 \cdot 11 && \text{Distribute the 4 times addend in the sum.} \\ &= 20 + 44 && \text{Multiply: } 4 \cdot 5 = 20 \text{ and } 4 \cdot 11 = 44. \\ &= 64 && \text{Add: } 20 + 44 = 64. \end{aligned}$$

Readers should check that the same answer is found by computing the sum within the parentheses first.

Exercise

Distribute: $5 \cdot (11 + 8)$.

Answer

95

The distributive property is the underpinning of the multiplication algorithm learned in our childhood years.

Example 9

Multiply: $6 \cdot 43$.

Solution

We'll express 43 as sum, then use the distributive property.

$$\begin{aligned} 6 \cdot 43 &= 6 \cdot (40 + 3) && \text{Express 43 as a sum: } 43 = 40 + 3 \\ &= 6 \cdot 40 + 6 \cdot 3 && \text{Distribute the 6.} \\ &= 240 + 18 && \text{Multiply: } 6 \cdot 40 = 240 \text{ and } 6 \cdot 3 = 18. \\ &= 258 && \text{Add: } 240 + 18 = 258. \end{aligned}$$

Readers should be able to see this application of the distributive property in the more familiar algorithmic form:

$$\begin{array}{r} 43 \\ \times 6 \\ \hline 18 \\ \underline{240} \\ 258 \end{array}$$

Or in the even more condensed form with “carrying:”

$$\begin{array}{r} ^1 43 \\ \times 6 \\ \hline 258 \end{array}$$

? Exercise

Use the distributive property to evaluate $8 \cdot 92$.

Answer

736

Multiplication is also distributive with respect to subtraction.

The Distributive Property (Subtraction)

Let a , b , and c be any whole numbers. Then,

$$a \cdot (b - c) = a \cdot b - a \cdot c.$$

We say the multiplication is “distributive with respect to subtraction.”

✓ Example 10

Use the distributive property to simplify: $3 \cdot (12 - 8)$.

Solution

This is the product of a number and a difference, so the distributive property may be applied.

$$\begin{aligned} 3 \cdot (12 - 8) &= 3 \cdot 12 - 3 \cdot 8 && \text{Distribute the 3 times each term in the difference.} \\ &= 36 - 24 && \text{Multiply: } 3 \cdot 12 = 36 \text{ and } 3 \cdot 8 = 24. \\ &= 12 && \text{Subtract: } 36 - 24 = 12. \end{aligned}$$

Alternate solution

Note what happens if we use the usual “order of operations” to evaluate the expression.

$$\begin{aligned} 3 \cdot (12 - 8) &= 3 \cdot 4 && \text{Parentheses first: } 12 - 8 = 4. \\ &= 12 && \text{Multiply: } 3 \cdot 4 = 12. \end{aligned}$$

Same answer.

? Exercise

Distribute: $8 \cdot (9 - 2)$.

Answer

56

Exercises

In Exercises 1-12, simplify the given expression.

1. $5+2 \cdot 2$
2. $5+2 \cdot 8$
3. $23 - 7 \cdot 2$
4. $37 - 3 \cdot 7$
5. $4 \cdot 3+2 \cdot 5$
6. $2 \cdot 5+9 \cdot 7$
7. $6 \cdot 5+4 \cdot 3$
8. $5 \cdot 2+9 \cdot 8$
9. $9+2 \cdot 3$
10. $3+6 \cdot 6$
11. $32 - 8 \cdot 2$
12. $24 - 2 \cdot 5$

In Exercises 13-28, simplify the given expression.

13. $45 \div 3 \cdot 5$
14. $20 \div 1 \cdot 4$
15. $2 \cdot 9 \div 3 \cdot 18$
16. $19 \cdot 20 \div 4 \cdot 16$
17. $30 \div 2 \cdot 3$
18. $27 \div 3 \cdot 3$
19. $8 - 6+1$
20. $15 - 5 + 10$
21. $14 \cdot 16 \div 16 \cdot 19$
22. $20 \cdot 17 \div 17 \cdot 14$
23. $15 \cdot 17 + 10 \div 10 - 12 \cdot 4$
24. $14 \cdot 18 + 9 \div 3 - 7 \cdot 13$
25. $22 - 10 + 7$
26. $29 - 11 + 1$
27. $20 \cdot 10 + 15 \div 5 - 7 \cdot 6$
28. $18 \cdot 19 + 18 \div 18 - 6 \cdot 7$

In Exercises 29-40, simplify the given expression.

29. $9+8 \div \{4+4\}$
30. $10 + 20 \div \{2+2\}$
31. $7 \cdot [8 - 5] - 10$
32. $11 \cdot [12 - 4] - 10$
33. $(18 + 10) \div (2 + 2)$
34. $(14 + 7) \div (2 + 5)$
35. $9 \cdot (10 + 7) - 3 \cdot (4 + 10)$
36. $9 \cdot (7 + 7) - 8 \cdot (3 + 8)$

37. $2 \cdot \{8 + 12\} \div 4$

38. $4 \cdot \{8+7\} \div 3$

39. $9+6 \cdot (12 + 3)$

40. $3+5 \cdot (10 + 12)$

In Exercises 41-56, simplify the given expression.

41. $2+9 \cdot [7 + 3 \cdot (9 + 5)]$

42. $6+3 \cdot [4 + 4 \cdot (5 + 8)]$

43. $7+3 \cdot [8 + 8 \cdot (5 + 9)]$

44. $4+9 \cdot [7 + 6 \cdot (3 + 3)]$

45. $6 - 5[11 - (2 + 8)]$

46. $15 - 1[19 - (7 + 3)]$

47. $11 - 1[19 - (2 + 15)]$

48. $9 - 8[6 - (2 + 3)]$

49. $4\{7[9 + 3] - 2[3 + 2]\}$

50. $4\{8[3 + 9] - 4[6 + 2]\}$

51. $9 \cdot [3 + 4 \cdot (5 + 2)]$

52. $3 \cdot [4 + 9 \cdot (8 + 5)]$

53. $3\{8[6 + 5] - 8[7 + 3]\}$

54. $2\{4[6 + 9] - 2[3 + 4]\}$

55. $3 \cdot [2 + 4 \cdot (9 + 6)]$

56. $8 \cdot [3 + 9 \cdot (5 + 2)]$

In Exercises 57-68, simplify the given expression.

57. $(5 - 2)^2$

58. $(5 - 3)^4$

59. $(4 + 2)^2$

60. $(3 + 5)^2$

61. $2^3 + 3^3$

62. $5^4 + 2^4$

63. $2^3 - 1^3$

64. $3^2 - 1^2$

65. $12 \cdot 5^2 + 8 \cdot 9+4$

66. $6 \cdot 3^2 + 7 \cdot 5 + 12$

67. $9 - 3 \cdot 2 + 12 \cdot 10^2$

68. $11 - 2 \cdot 3 + 12 \cdot 4^2$

In Exercises 69-80, simplify the given expression.

69. $4^2 - (13 + 2)$

70. $3^3 - (7 + 6)$

71. $3^3 - (7 + 12)$

72. $4^3 - (6 + 5)$

73. $19 + 3[12 - (2^3 + 1)]$

74. $13 + 12[14 - (2^2 + 1)]$

75. $17 + 7[13 - (2^2 + 6)]$

76. $10 + 1[16 - (2^2 + 9)]$

77. $4^3 - (12 + 1)$

78. $5^3 - (17 + 15)$

79. $5 + 7[11 - (2^2 + 1)]$

80. $10 + 11[20 - (2^2 + 1)]$

In Exercises 81-92, simplify the given expression.

81. $\frac{13+35}{3(4)}$

82. $\frac{35+28}{7(3)}$

83. $\frac{64-(8\cdot6-3)}{4\cdot7-9}$

84. $\frac{19-(4\cdot3-2)}{6\cdot3-9}$

85. $\frac{2+13}{4-1}$

86. $\frac{7+1}{8-4}$

87. $\frac{17+14}{9-8}$

88. $\frac{16+2}{13-11}$

89. $\frac{37+27}{8(2)}$

90. $\frac{16+38}{6(3)}$

91. $\frac{40-(3\cdot7-9)}{8\cdot2-2}$

92. $\frac{60-(8\cdot6-3)}{5\cdot4-5}$

In Exercises 93-100, use the distributive property to evaluate the given expression.

93. $5 \cdot (8 + 4)$

94. $8 \cdot (4 + 2)$

95. $7 \cdot (8 - 3)$

96. $8 \cdot (9 - 7)$

97. $6 \cdot (7 - 2)$

98. $4 \cdot (8 - 6)$

99. $4 \cdot (3 + 2)$

100. $4 \cdot (9 + 6)$

In Exercises 101-104, use the distributive property to evaluate the given expression using the technique shown in Example 9.

101. $9 \cdot 62$

102. $3 \cdot 76$

103. $3 \cdot 58$

104. $7 \cdot 57$

Answers

1. 9
3. 9
5. 22
7. 42
9. 15
11. 16
13. 75
15. 108
17. 45
19. 3
21. 266
23. 208
25. 19
27. 161
29. 10
31. 11
33. 7
35. 111
37. 10
39. 99
41. 443
43. 367
45. 1
47. 9
49. 296
51. 279
53. 24
55. 186
57. 9
59. 36
61. 35
63. 7
65. 376
67. 1203
69. 1
71. 8
73. 28

75. 38
77. 51
79. 47
81. 4
83. 1
85. 5
87. 31
89. 4
91. 2
93. 60
95. 35
97. 30
99. 20
101. 558
103. 174

¹Later, we'll see that this property applies to all numbers, not just whole numbers

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CHAPTER OVERVIEW

2: Fractions

In this chapter, you'll learn how we use fractions.

[2.1: Prelude to Fractions - Egyptian Math](#)

[2.2: Equivalent Fractions](#)

[2.3: Multiplying Fractions](#)

[2.4: Dividing Fractions](#)

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[2.6: Multiplying and Dividing Mixed Fractions](#)

[2.7: Adding and Subtracting Mixed Fractions](#)

[2.8: Order of Operations with Fractions](#)

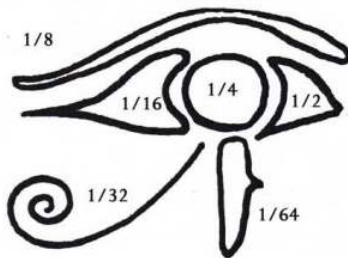
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2.1: Prelude to Fractions - Egyptian Math

Around 3000 BC, Egyptians were carving hieroglyphs into stone monuments to their kings and queens. Hieroglyphs are pictures that represent objects and they were used for words and numbers. Oddly, fractions were always written as sums of “unit fractions,” fractions whose numerator is always 1. For instance, instead of writing $\frac{3}{5}$, they would write a sum of unit fractions.

$$\frac{3}{5} = \frac{1}{2} + \frac{1}{10}$$

Much of the ancient Egyptian math that we know of was in service to the agricultural and economic life of the people. In measuring dry goods such as grains, special glyphs were used to represent basic fractional amounts, glyphs that came together to represent the Eye of Horus.



Horus was a falcon-god whose father Osiris was murdered by his own brother Seth. When Horus attempted to avenge his father's death, Seth ripped out Horus' eye and cut it into six pieces, scattering them throughout Egypt.

Taking pity on Horus, Thot, the god of learning and magic, found the pieces and put them back together making Horus healthy and whole again. Each piece of the Eye of Horus represents a different fraction of a hekat, or volume of grain. It was written that an apprentice scribe added the fractions one day and got

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} = \frac{63}{64}.$$

Asking where the missing $\frac{1}{64}$ was, he was told that Thot would make up the difference to anyone “who sought and accepted is protection.”

In this chapter, you'll learn how we use fractions.

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2.2: Equivalent Fractions

In this section we deal with fractions, numbers or expressions of the form a/b .

Definition: Fractions

A number of the form

$$\frac{a}{b}$$

where a and b are numbers is called a *fraction*. The number a is called the *numerator* of the fraction, while the number b is called the *denominator* of the fraction.

Near the end of this section, we'll see that the numerator and denominator of a fraction can also be algebraic expressions, but for the moment we restrict our attention to fractions whose numerators and denominators are integers. We start our study of fractions with the definition of *equivalent fractions*.

Equivalent Fractions

Two fractions are *equivalent* if they represent the same numerical value.

But how can we tell if two fractions represent the same number? Well, one technique involves some simple visualizations. Consider the image shown in Figure 4.1, where the shaded region represents $1/3$ of the total area of the figure (one of three equal regions is shaded).



Figure 4.1: The shaded region is $1/3$ of the whole region.

In Figure 4.2, we've shaded $2/6$ of the entire region (two of six equal regions are shaded).



Figure 4.2: The shaded region is $2/6$ of the whole region.

In Figure 4.3, we've shaded $4/12$ of the entire region (four of twelve equal regions are shaded).



Figure 4.3: The shaded region is $4/12$ of the whole region.

Let's take the diagrams from Figure 4.1, Figure 4.2, and Figure 4.3 and stack them one atop the other, as shown in Figure 4.4.

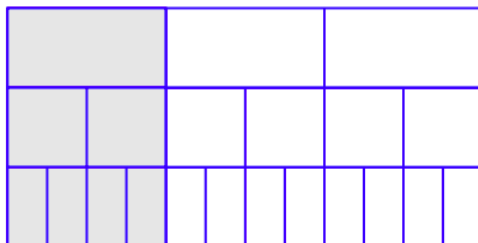


Figure 4.4: One of three equals two of six equals four of twelve.

Figure 4.4 provides solid visual evidence that the following fractions are equivalent.

$$\frac{1}{3} = \frac{2}{6} = \frac{4}{12}$$

Key Observations

1. If we start with the fraction $\frac{1}{3}$, then multiply both numerator and denominator by 2, we get the following result.

$$\begin{aligned}\frac{1}{3} &= \frac{1 \cdot 2}{3 \cdot 2} && \text{Multiply the numerator and denominator by 2.} \\ &= \frac{2}{6} && \text{Simplify numerator and denominator.}\end{aligned}$$

This is precisely the same thing that happens going from Figure 4.1 to 4.2, where we double the number of available boxes (going from 3 available to 6 available) and double the number of shaded boxes (going from 1 shaded to 2 shaded).

2. If we start with the fraction $\frac{1}{3}$, then multiply both numerator and denominator by 4, we get the following result.

$$\begin{aligned}\frac{1}{3} &= \frac{1 \cdot 4}{3 \cdot 4} && \text{Multiply numerator and denominator by 4.} \\ &= \frac{4}{12} && \text{Simplify numerator and denominator.}\end{aligned}$$

This is precisely the same thing that happens going from Figure 4.1 to 4.3, where we multiply the number of available boxes by 4 (going from 3 available to 12 available) and multiply the number of shaded boxes by 4 (going from 1 shaded to 4 shaded).

The above discussion motivates the following fundamental result.

Creating Equivalent Fractions

If you start with a fraction, then multiply both its numerator and denominator by the same number, the resulting fraction is equivalent (has the same numerical value) to the original fraction. In symbols,

$$\frac{a}{b} = \frac{a \cdot x}{b \cdot x}$$

Arguing in Reverse

Reversing the above argument also holds true.

1. If we start with the fraction $\frac{2}{6}$, then divide both numerator and denominator by 2, we get the following result.

$$\begin{aligned}\frac{2}{6} &= \frac{2 \div 2}{6 \div 2} && \text{Divide numerator and denominator by 2.} \\ &= \frac{1}{3} && \text{Simplify numerator and denominator.}\end{aligned}$$

This is precisely the same thing that happens going backwards from Figure 4.2 to 4.1, where we divide the number of available boxes by 2 (going from 6 available to 3 available) and dividing the number of shaded boxes by 2 (going from 2 shaded to 1 shaded).

2. If we start with the fraction $\frac{4}{12}$, then divide both numerator and denominator by 4, we get the following result.

$$\begin{aligned}\frac{4}{12} &= \frac{4 \div 4}{12 \div 4} && \text{Divide numerator and denominator by 4.} \\ &= \frac{1}{3} && \text{Simplify numerator and denominator.}\end{aligned}$$

This is precisely the same thing that happens going backwards from Figure 4.3 to 4.1, where we divide the number of available boxes by 4 (going from 12 available to 3 available) and divide the number of shaded boxes by 4 (going from 4 shaded to 1 shaded).

The above discussion motivates the following fundamental result.

Creating Equivalent Fractions

If you start with a fraction, then divide both its numerator and denominator by the same number, the resulting fraction is equivalent (has the same numerical value) to the original fraction. In symbols,

$$\frac{a}{b} = \frac{a \div x}{b \div x}$$

The Greatest Common Divisor

We need a little more terminology.

Divisor

If d and a are natural numbers, we say that “ d divides a ” if and only if when a is divided by d , the remainder is zero. In this case, we say that “ d is a divisor of a .”

For example, when 36 is divided by 4, the remainder is zero. In this case, we say that “4 is a divisor of 36.” On the other hand, when 25 is divided by 4, the remainder is not zero. In this case, we say that “4 is not a divisor of 25.”

Greatest Common Divisor

Let a and b be natural numbers. The common divisors of a and b are those natural numbers that divide both a and b . The *greatest common divisor* is the largest of these common divisors.

✓ Example 1

Find the greatest common divisor of 18 and 24.

Solution

First list the divisors of each number, the numbers that divide each number with zero remainder.

Divisors of 18 : 1, 2, 3, 6, 9, and 18

Divisors of 24 : 1, 2, 3, 4, 6, 8, 12, and 24

The common divisors are:

Common Divisors : 1, 2, 3, and 6

The greatest common divisor is the largest of the common divisors. That is,

Greatest Common Divisor = 6.

That is, the largest number that divides both 18 and 24 is the number 6.

? Exercise

Find the greatest common divisor of 12 and 18.

Answer

6

Reducing a Fraction to Lowest Terms

First, a definition.

Lowest Terms

A fraction is said to be *reduced to lowest terms* if the greatest common divisor of both numerator and denominator is 1.

Thus, for example, $\frac{2}{3}$ is reduced to lowest terms because the greatest common divisor of 2 and 3 is 1. On the other hand, $\frac{4}{6}$ is **not** reduced to lowest terms because the greatest common divisor of 4 and 6 is 2.

✓ Example 2

Reduce the fraction 18/24 to lowest terms.

Solution

One technique that works well is dividing both numerator and denominator by the greatest common divisor of the numerator and denominator. In Example 1, we saw that the greatest common divisor of 18 and 24 is 6. We divide both numerator and denominator by 6 to get

$$\begin{aligned}\frac{18}{24} &= \frac{18 \div 6}{24 \div 6} && \text{Divide numerator and denominator by 6.} \\ &= \frac{3}{4} && \text{Simplify numerator and dice.}\end{aligned}$$

Note that the greatest common divisor of 3 and 4 is now 1. Thus, 3/4 is reduced to lowest terms.

There is a second way we can show division of numerator and denominator by 6. First, factor both numerator and denominator as follows:

$$\frac{18}{24} = \frac{3 \cdot 6}{4 \cdot 6} \quad \text{Factor out a 6.}$$

You can then show “division” of both numerator and denominator by 6 by “crossing out” or “canceling” a 6 in the numerator for a 6 in the denominator, like this:

$$\begin{aligned}&= \frac{3 \cdot \cancel{6}}{4 \cdot \cancel{6}} && \text{Cancel common factor.} \\ &= \frac{3}{4}\end{aligned}$$

Note that we get the same equivalent fraction, reduced to lowest terms, namely 3/4.

? Exercise

Reduce the fraction 12/18 to lowest terms.

Answer

2/3

📌 Important Point

In Example 2 we saw that 6 was both a *divisor* and a *factor* of 18. The words *divisor* and *factor* are equivalent.

We used the following technique in our second solution in Example 2.

✎ Cancellation Rule

If you express numerator and denominator as a product, then you may cancel common factors from the numerator and denominator. The result will be an equivalent fraction.

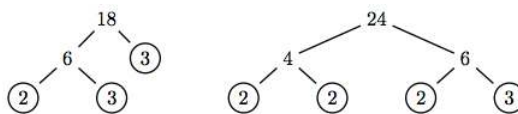
Because of the “Cancellation Rule,” one of the most effective ways to reduce a fraction to lowest terms is to first find prime factorizations for both numerator and denominator, then cancel all common factors.

✓ Example 3

Reduce the fraction 18/24 to lowest terms.

Solution

Use factor trees to prime factor numerator and denominator.



Once we've factored the numerator and denominator, we cancel common factors.

$$\begin{aligned}\frac{18}{24} &= \frac{2 \cdot 3 \cdot 3}{2 \cdot 2 \cdot 2 \cdot 3} && \text{Prime factor numerator and denominator.} \\ &= \frac{\cancel{2} \cdot \cancel{3} \cdot 3}{\cancel{2} \cdot 2 \cdot 2 \cdot \cancel{3}} && \text{Cancel common factors.} \\ &= \frac{3}{2 \cdot 2} && \text{Remaining factors.} \\ &= \frac{3}{4} && \text{Simplify denominator.}\end{aligned}$$

Thus, $18/24 = 3/4$.

? Exercise 2.2.1

Reduce the fraction $28/35$ to lowest terms.

Answer

$4/5$

✓ Example 4

Reduce the fraction $28/42$ to lowest terms.

Solution

Use factor trees to prime factor numerator and denominator.



Now we can cancel common factors.

$$\begin{aligned}\frac{28}{42} &= \frac{2 \cdot 2 \cdot 7}{2 \cdot 3 \cdot 7} && \text{Prime factor numerator and denominator.} \\ &= \frac{\cancel{2} \cdot 2 \cdot \cancel{7}}{\cancel{2} \cdot 3 \cdot \cancel{7}} && \text{Cancel common factors.} \\ &= \frac{2}{3}\end{aligned}$$

Thus, $28/42 = 2/3$.

? Exercise

Reduce the fraction $36/60$ to lowest terms.

Answer

$3/5$

Reducing Fractions with Variables

We use exactly the same technique to reduce fractions whose numerators and denominators contain variables.

✓ Example 5

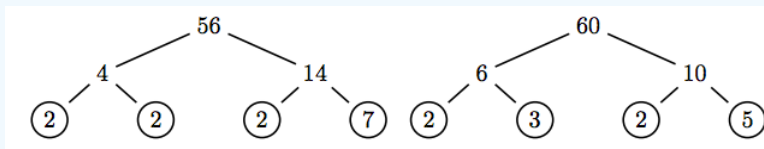
Reduce

$$\frac{56x^2y}{60xy^2}$$

to lowest terms.

Solution

Use factor trees to factor the coefficients of numerator and denominator.



Now cancel common factors.

$$\begin{aligned}
 \frac{56x^2y}{60xy^2} &= \frac{2 \cdot 2 \cdot 2 \cdot 7 \cdot x \cdot x \cdot y}{2 \cdot 2 \cdot 3 \cdot 5 \cdot x \cdot y \cdot y} && \text{Prime factor numerator and denominator.} \\
 &= \frac{\cancel{2} \cdot \cancel{2} \cdot 2 \cdot 7 \cdot \cancel{x} \cdot x \cdot \cancel{y}}{\cancel{2} \cdot \cancel{2} \cdot 3 \cdot 5 \cdot \cancel{x} \cdot y \cdot \cancel{y}} && \text{Cancel common factors.} \\
 &= \frac{2 \cdot 7 \cdot x}{3 \cdot 5 \cdot y} && \text{Remaining factors.} \\
 &= \frac{14x}{15y} && \text{Simplify numerator and denominator.}
 \end{aligned}$$

Thus, $56x^2y/(60xy^2) = 14x/(15y)$.

? Exercise

Reduce:

$$\frac{25a^3b}{40a^2b^3}$$

Answer

$$\frac{5a}{8b^2}$$

A Word on Mathematical Notation

There are two types of mathematical notation: (1) inline mathematical notation, and (2) displayed mathematical notation.

Inline Mathematical Notation

The notation $14x/(15y)$ is called **inline mathematical notation**. When the same expression is centered on its own line, as in

$$\frac{14x}{15y},$$

this type of notation is called displayed mathematical notation.

When you work a problem by hand, using pencil and paper calculations, the preferred format is displayed notation, like the displayed notation used to simplify the given expression in Example 5. However, computers and calculators require that you enter your expressions using inline mathematical notation. Therefore, it is extremely important that you are equally competent with either mathematical notation: displayed or inline.

By the way, order of operations, when applied to the inline expression $14x/(15y)$, requires that we perform the multiplication inside the parentheses first. Then we must perform multiplications and divisions as they occur, as we move from left to right through the expression. This is why the inline notation $14x/(15y)$ is equivalent to the displayed notation

$$\frac{14x}{15y}.$$

However, the expression $14x/15y$ is a different beast. There are no parentheses, so we perform multiplication and division as they occur, moving left to right through the expression. Thus, we must first take the product of 14 and x, divide the result by 15, then multiply by y. In displayed notation, this result is equivalent to

$$\frac{14x}{15} \cdot y,$$

which is a different result.

Some readers might wonder why we did not use the notation $(14x)/(15y)$ to describe the solution in Example 5. After all, this inline notation is also equivalent to the displayed notation

$$\frac{14x}{15y}.$$

However, the point is that we do not need to, as order of operations already requires that we take the product of 14 and x before dividing by 15y. If this is hurting your head, know that it's quite acceptable to use the equivalent notation $(14x)/(15y)$ instead of $14x/(15y)$. Both are correct.

Equivalent Fractions in Higher Terms

Sometimes the need arises to find an equivalent fraction with a different, larger denominator.

✓ Example 6

Express $3/5$ as an equivalent fraction having denominator 20.

Solution

The key here is to remember that multiplying numerator and denominator by the same number produces an equivalent fraction. To get an equivalent fraction with a denominator of 20, we'll have to multiply numerator and denominator of $3/5$ by 4.

$$\begin{aligned} & \frac{3}{5} \quad \text{Multiply numerator and denominator by 4.} \\ = & \frac{12}{20} \quad \text{Simplify numerator and denominator.} \end{aligned}$$

Therefore, $3/5$ equals $12/20$.

? Exercise

Express $2/3$ as an equivalent fraction having denominator 21.

Answer

$14/21$

✓ Example 7

Express 8 as an equivalent fraction having denominator 5.

Solution

The key here is to note that

$$8 = \frac{8}{1} \quad \text{Understood denominator is 1.}$$

To get an equivalent fraction with a denominator of 5, we'll have to multiply numerator and denominator of 8/1 by 5.

$$\begin{aligned} &= \frac{8 \cdot 5}{1 \cdot 5} \quad \text{Multiply numerator and denominator by 5.} \\ &= \frac{40}{5} \quad \text{Simplify numerator and denominator.} \end{aligned}$$

Therefore, 8 equals 40/5.

? Exercise

Express 5 as an equivalent fraction having denominator 7.

Answer

$$35/7$$

✓ Example 8

Express 2/9 as an equivalent fraction having denominator 18a.

Solution

To get an equivalent fraction with a denominator of 18a, we'll have to multiply numerator and denominator of 2/9 by 2a.

$$\begin{aligned} \frac{2}{9} &= \frac{2 \cdot 2a}{9 \cdot 2a} \quad \text{Multiply numerator and denominator by } 2a. \\ &= \frac{4a}{18a} \quad \text{Simplify numerator and denominator.} \end{aligned}$$

Therefore, 2/9 equals 4a/(18a), or equivalently, (4a)/(18a).

? Exercise

Express 3/8 as an equivalent fraction having denominator 24a.

Answer

$$\frac{9a}{24a}$$

Negative Fractions

We have to also deal with fractions that are negative. First, let's discuss placement of the negative sign.

- Positive divided by negative is negative, so

$$\frac{3}{-5} = -\frac{3}{5}.$$

- But it is also true that negative divided by positive is negative. Thus,

$$\frac{-3}{5} = -\frac{3}{5}.$$

These two observations imply that all three of the following fractions are equivalent (the same number):

$$\frac{3}{-5} = -\frac{3}{5} = \frac{-3}{5}.$$

Note that there are three possible placements for the negative sign: (1) the denominator, (2) the fraction bar, or (3) the numerator. Any one of these placements produces an equivalent fraction.

Fractions and Negative Signs

Let a and b be any integers. All three of the following fractions are equivalent (same number):

$$\frac{a}{-b} = -\frac{a}{b} = \frac{-a}{b}.$$

Mathematicians prefer to place the negative sign either in the numerator or on the fraction bar. The use of a negative sign in the denominator is discouraged.

✓ Example 9

Reduce:

$$\frac{50x^3}{-75x^5}$$

to lowest terms.

Solution

Prime factor numerator and denominator and cancel.

$$\begin{aligned} \frac{50x^3}{-75x^5} &= \frac{2 \cdot 5 \cdot 5 \cdot x \cdot x \cdot x}{-3 \cdot 5 \cdot 5 \cdot x \cdot x \cdot x \cdot x \cdot x} \\ &= \frac{2 \cdot \cancel{5} \cdot \cancel{5} \cdot \cancel{x} \cdot \cancel{x} \cdot \cancel{x}}{-3 \cdot \cancel{5} \cdot \cancel{5} \cdot \cancel{x} \cdot \cancel{x} \cdot \cancel{x} \cdot x \cdot x} \\ &= \frac{2}{-3 \cdot x \cdot x} \\ &= \frac{2}{-3x^2} \end{aligned}$$

However, it is preferred that there be no negative signs in the denominator, so let's place the negative sign on the fraction bar (the numerator would suit as well). Thus,

$$\frac{50x^3}{-75x^5} = -\frac{2}{3x^2}$$

We also have the following result.

Fractions and Negative Signs

Let a and b be any integers. Then,

$$\frac{-a}{-b} = \frac{a}{b}.$$

✓ Example 10

Reduce:

$$\frac{-12xy^2}{-18x^2y}$$

Solution

Unlike Example 9, some like to take care of the sign of the answer first.

$$\frac{-12xy^2}{-18x^2y} = \frac{12xy^2}{18x^2y}$$

Now we can factor numerator and denominator and cancel common factors.

$$\begin{aligned} &= \frac{2 \cdot 2 \cdot 3 \cdot x \cdot y \cdot y}{2 \cdot 3 \cdot 3 \cdot x \cdot x \cdot y} \\ &= \frac{\cancel{2} \cdot \cancel{2} \cdot \cancel{3} \cdot \cancel{x} \cdot y \cdot \cancel{y}}{\cancel{2} \cdot \cancel{3} \cdot 3 \cdot \cancel{x} \cdot x \cdot \cancel{y}} = \frac{2y}{3x} \end{aligned}$$

Thus,

$$\frac{-12xy^2}{-18x^2y} = \frac{2y}{3x}.$$

? Exercise

Reduce:

$$\frac{-21a^2b^3}{-56a^3b}$$

Answer

$$\frac{3b^2}{8a}$$

Exercises

In Exercises 1-12, find the GCD of the given numbers.

1. 72, 8
2. 76, 52
3. 52, 20
4. 56, 96
5. 36, 63
6. 63, 21
7. 72, 44
8. 10, 40
9. 16, 56
10. 54, 66
11. 84, 24
12. 75, 45

In Exercises 13-28, reduce the given fraction to lowest terms.

13. $\frac{22}{98}$
14. $\frac{28}{56}$
15. $\frac{93}{15}$

16. $\frac{90}{39}$

17. $\frac{69}{21}$

18. $\frac{74}{62}$

19. $\frac{74}{12}$

20. $\frac{66}{10}$

21. $\frac{66}{57}$

22. $\frac{34}{30}$

23. $\frac{33}{99}$

24. $\frac{20}{58}$

25. $\frac{69}{24}$

26. $\frac{18}{96}$

27. $\frac{46}{44}$

28. $\frac{92}{24}$

29. Express 3 as an equivalent fraction having denominator 24. 30. Express 3 as an equivalent fraction having denominator 8. 31. Express $\frac{25}{19}$ as an equivalent fraction having denominator 57. 32. Express $\frac{29}{22}$ as an equivalent fraction having denominator 44. 33. Express 2 as an equivalent fraction having denominator 2. 34. Express 2 as an equivalent fraction having denominator 8. 35. Express $\frac{18}{19}$ as an equivalent fraction having denominator 95. 36. Express $\frac{17}{22}$ as an equivalent fraction having denominator 44. 37. Express $\frac{1}{3}$ as an equivalent fraction having denominator 24. 38. Express $\frac{15}{19}$ as an equivalent fraction having denominator 95. 39. Express 16 as an equivalent fraction having denominator 4. 40. Express 5 as an equivalent fraction having denominator 2.

In Exercises 41-56, reduce the given fraction to lowest terms.

41. $\frac{34}{-86}$

42. $\frac{-48}{14}$

43. $\frac{-72}{-92}$

44. $\frac{27}{-75}$

45. $\frac{-92}{82}$

46. $\frac{-44}{-62}$

47. $\frac{-21}{33}$

48. $\frac{57}{-99}$

49. $\frac{22}{-98}$

50. $\frac{-33}{69}$

51. $\frac{42}{-88}$

52. $\frac{-100}{48}$

53. $\frac{94}{-6}$

54. $\frac{-36}{-38}$

55. $\frac{10}{-86}$

56. $\frac{-100}{-46}$

57. Express $\frac{3}{2}$ as an equivalent fraction having denominator 62n.

58. Express $\frac{6}{25}$ as an equivalent fraction having denominator 50a.

59. Express $\frac{13}{10}$ as an equivalent fraction having denominator 60m.

60. Express $\frac{1}{16}$ as an equivalent fraction having denominator 80p.

61. Express $\frac{3}{2}$ as an equivalent fraction having denominator 50n.

62. Express $\frac{43}{38}$ as an equivalent fraction having denominator 76a.

63. Express 11 as an equivalent fraction having denominator 4m. 64. Express 13 as an equivalent fraction having denominator 6n.

65. Express 3 as an equivalent fraction having denominator 10m.

66. Express 10 as an equivalent fraction having denominator 8b.

67. Express 6 as an equivalent fraction having denominator 5n.

68. Express 16 as an equivalent fraction having denominator 2y.

In Exercises 69-84, reduce the given fraction to lowest terms.

69. $\frac{82y^5}{-48y}$

70. $\frac{-40y^5}{-55y}$

71. $\frac{-77x^5}{44x^4}$

72. $\frac{-34x^6}{-80x}$

73. $\frac{-14y^5}{54y^2}$

74. $\frac{96y^4}{-40y^2}$

75. $\frac{42x}{81x^3}$

76. $\frac{26x^2}{32x^6}$

77. $\frac{-12x^5}{14x^6}$

78. $\frac{-28y^4}{72y^6}$

79. $\frac{-74x}{22x^2}$

80. $\frac{56x^2}{26x^3}$

81. $\frac{-12y^5}{98y^6}$

82. $\frac{96x^2}{14x^4}$

83. $\frac{18x^6}{-54x^2}$

84. $\frac{32x^6}{62x^2}$

In Exercises 85-100, reduce the given fraction to lowest terms.

85. $\frac{26y^2x^4}{-62y^6x^2}$

86. $\frac{6x^2y^3}{40x^3y^2}$

87. $\frac{-2y^6x^4}{-94y^2x^5}$

88. $\frac{90y^6x^3}{39y^3x^5}$

89. $\frac{30y^5x^5}{-26yx^4}$

90. $\frac{74x^6y^4}{-52xy^3}$

91. $\frac{36x^3y^2}{-98x^4y^5}$

92. $\frac{84x^3y}{16x^4y^2}$

93. $\frac{-8x^6y^3}{54x^3y^5}$

94. $\frac{70y^5x^2}{16y^4x^5}$

95. $\frac{34yx^6}{-58y^5x^4}$

96. $\frac{99y^2x^3}{88y^6x}$

97. $\frac{-36y^3x^5}{51y^2x}$

98. $\frac{44y^5x^5}{-88y^4x}$

99. $\frac{91y^3x^2}{-28y^5x^5}$

100. $\frac{-76y^2x}{-57y^5x^6}$

101. **Hurricanes.** According to the National Atmospheric and Oceanic Administration, in 2008 there were 16 named storms, of which 8 grew into hurricanes and 5 were major.

- i) What fraction of named storms grew into hurricanes? Reduce your answer to lowest terms.
- ii) What fraction of named storms were major hurricanes? Reduce your answer to lowest terms.
- iii) What fraction of hurricanes were major? Reduce your answer to lowest terms.

102. **Tigers.** Tigers are in critical decline because of human encroachment, the loss of more than nine-tenths of their habitat, and the growing trade in tiger skins and body parts. *Associated Press-Times-Standard 01/24/10 Pressure mounts to save the tiger.*

- i) Write the loss of habitat as a fraction.

- ii) Describe in words what the numerator and denominator of this fraction represent.
- iii) If the fraction represents the loss of the whole original habitat, how much of the original habitat remains?
-

Answers

1. 8

3. 4

5. 9

7. 4

9. 8

11. 12

13. $\frac{11}{49}$

15. $\frac{31}{5}$

17. $\frac{23}{7}$

19. $\frac{37}{6}$

21. $\frac{22}{19}$

23. $\frac{1}{3}$

25. $\frac{23}{8}$

27. $\frac{23}{22}$

29. $\frac{72}{24}$

31. $\frac{75}{57}$

33. $\frac{4}{2}$

35. $\frac{90}{95}$

37. $\frac{8}{24}$

39. $\frac{64}{4}$

41. $\frac{-17}{43}$

43. $\frac{18}{23}$

45. $\frac{-46}{41}$

47. $\frac{-7}{11}$

49. $\frac{-11}{49}$

51. $\frac{-21}{44}$

53. $\frac{-47}{3}$

55. $\frac{-5}{43}$

57. $\frac{93n}{62n}$

59. $\frac{78m}{60m}$

61. $\frac{75n}{50n}$

63. $\frac{44m}{4m}$

65. $\frac{30m}{10m}$

67. $\frac{30n}{5n}$

69. $\frac{-41y^4}{24}$

71. $\frac{-7x}{4}$

73. $-\frac{7y^3}{27}$

75. $\frac{14}{27x^2}$

77. $-\frac{6}{7x}$

79. $-\frac{37}{11x}$

81. $-\frac{6}{49y}$

83. $-\frac{x^4}{3}$

85. $-\frac{13x^2}{31y^4}$

87. $\frac{y^4}{47x}$

89. $-\frac{15y^4x}{13}$

91. $-\frac{18}{49xy^3}$

93. $-\frac{4x^3}{27y^2}$

95. $-\frac{17x^2}{29y^4}$

97. $-\frac{12yx^4}{17}$

99. $-\frac{13}{4y^2x^3}$

101.

i) $\frac{1}{2}$

ii) $\frac{5}{16}$

iii) $\frac{5}{8}$

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